

Chapter 4

Genericity and habituality in Czech

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In Slavic languages, the domain of genericity (and habituality) is traditionally viewed as associated with contextually determined meanings of imperfective forms, and to a lesser extent of perfective forms. What has so far remained largely unexplored, and misunderstood, is the verb marker *-va-*, which is dedicated to deriving verbs that only have a generic (and habitual) interpretation. Based mainly on Czech data, it will be shown that the verb marker *-va-* has distributional and semantic properties that preempt its analysis as a marker of tense and/or aspect (*pace* Dahl 1995, i.a.). Neither can its meaning be reduced to standard (modal) operators that are used in the analysis of progressive or imperfective aspect (*pace* suggestions in Cable 2022, i.a.). Its quantificational and modal (intensional) components strongly suggest that it is a type of generic marker delimiting a subclass of generic sentence. The insights reached here support the view that generic markers in natural languages not only exist, but are not as rare as is generally thought, and by the same token also bolster independent arguments made elsewhere for the independence of genericity (and habituality) from the categories of the TMA (tense, modality, aspect) system.

1 Main data and questions

Generic sentences are sentences like *Dogs bark*, *Sugar dissolves in water*, *Sam walks to school on Mondays*. This chapter focuses on their encoding in Slavic languages that have a dedicated marker on the verb enforcing only a generic (or habitual) reading of a sentence, besides using imperfective, and also perfective, verbs for this purpose in an appropriate context that triggers their generic (or habitual) interpretation. Its instantiation in Czech (West Slavic language) is treated as a suffix in standard reference grammars (see Petr 1986: 184, Dahl 1995, Kosek



2014, Nübler 2017, for instance); some Bohemists treat it as an infix (e.g., Kučera 1981, Bílý 1986). In Genis et al. (2021: 23), it is treated as an infix across different Slavic languages.¹ Henceforth, I will use *-va-* as its citation form (as is also common in Bohemistics), and refer to it as the MORPHEME *-va-* OR MARKER *-va-*, given that trying to resolve the suffix-infix controversy would go well beyond the scope of this contribution.

This marker is attested in all Slavic languages (see Genis et al. 2021 for examples from different Slavic languages). However, it is productive only in West Slavic languages, specifically in Czech, Slovak and to a lesser extent in Polish, and commonly used in both spoken and written language (see e.g., Širokova 1963: 62, Comrie 1976: 27, Kučera 1981, Bílý 1985, Bílý 1986, Petr 1986, Genis et al. 2021, among others).² In East and South Slavic languages it is also attested, but it is reported to be largely unproductive, at least in their standard varieties.³

Here, the main focus is on the instantiation of this Slavic marker in Czech. The constraints on the length of this chapter prohibit me from adducing relevant examples from other Slavic languages. To the extent that the relevant generic verb forms exist in other Slavic languages, e.g., *pisývat* in Russian, meaning ‘to write repeatedly, sporadically, as a rule’ (albeit judged as archaic), it is reasonable to expect that the arguments made here for the corresponding Czech forms, e.g., *psávat*, should apply to them, as well.

Let me start with a few examples, given below:

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|-----|----|---|------|
| (1) | a. | <i>Psi štěkají.</i>
dogs bark
‘Dogs bark.’ [Intended interpretation: ‘Dogs as a kind have the characterizing property of barking.’] | IPFV |
|-----|----|---|------|

¹Genis et al. (2021: 23): “[T]he most common infix in this [habitual, frequentative/iterative, HF] function across languages seems to be one that might be rendered *-va/-*.”

²The productivity of *-va-* in Czech is also evident in the observation that children use verbs on which it occurs early in their language acquisition, even before they master the grammatical perfective/imperfective distinction (Chmelíčková 2005).

³In standard spoken and written Russian, the generic marker only occurs on some verbs that are taken to be marginal in use (Comrie 1976: fn. 1 on p. 27, who cites Isačenko 1962: 405–407). They are attested in fixed collocations (e.g., *V žizni tak byvaet* [gloss: in life so be.GEN.3.SG.PRS] ‘That’s how life goes. / That’s life.’). Dickey (2000: 87) claims that in standard Serbian explicitly “habitual” (in his terms, generic in my terms) verbs can be formed with the morpheme *-va-* (which is an infix, according to him) from biaspectual verbs, e.g.: *kršćavati* (from *krstiti*^{PFV/IPFV} ‘baptize’); *noćivati* (from *noćiti*^{PFV/IPFV} ‘spend the night’); *ručavati* (from *ručati*^{PFV/IPFV} ‘eat lunch’); *večeravati* (from *večerati*^{PFV/IPFV} ‘eat dinner’). However, Dickey’s claim is contested in Genis et al. (2021: 26): “[A]ccording to our observations, these verbs do not always have a habitual interpretation but can be used in a durative meaning, as well; other than their base verbs they are explicitly imperfective.”

- b. Psi se *poštěkají*, že ani nevíte, jak k tomu došlo. PFV
 dogs REFL PREF.bark that even NEG.know how to it came
 ‘Dogs (will) bark (out) at each other, and one hardly knows why.’
- c. Psi *poštěkávají* ve spánku, vrčí a cukají packami. IPFV
 dogs PREF.bark.IPFV in sleep, growl and twitch paws
 ‘Dogs bark now and then while sleeping, growl and twitch their paws.’ [i.e., bark a little/on and off]
- d. Psi *štěkávají* na ty, které neznají. GEN
 dogs bark.GEN at those whom NEG.know
 ‘Dogs bark at those whom they don’t know.’

As the English translations of the four Czech generic sentences above show, one and the same English verb form *bark* (3rd person plural, present tense) corresponds to four different verb forms in Czech. All four share the same root *-štěk-* ‘bark’. (1a) is headed by a primary (root) imperfective verb, formally unmarked for imperfectivity, while (1c) by a secondary imperfective verb, which is derived by means of the imperfectivizing suffix (glossed with IPFV) from a perfective base, realized in the perfective verb in (1b). In languages with the perfective/imperfective distinction like Slavic or Romance languages, the semantics of imperfectivity is taken to have close affinities to genericity (and habituality), based on the observation that imperfective verbs have common contextually determined generic readings, as we see in (1a) and (1c).

However, there is no incompatibility between the semantics of perfectivity and genericity (and habituality). Perfective verbs are also used in generic sentences, and this use is not as unusual or rare as some assume (*pace* Comrie 1976, Dahl 1995, i.a.). For instance, in Romance languages, and in West Slavic languages (see e.g., Filip & Carlson 1997), perfective verbs have common contextually determined generic readings, as is illustrated by (1b). Here, we have a perfective verb derived by means of the prefix *po-* from the stem of the primary (root) imperfective verb *štěkat* ‘to (be) bark(ing)’ (used in (1a)).⁴

Most importantly, (1d) has an obligatory generic interpretation in all contexts, and unlike the (im)perfective verb forms in (1a) and (1c), (1d) can never be used in episodic contexts. This is due to the morpheme *-va-* on its main verb. It constitutes a sufficient, but not a necessary, condition for the encoding of generic statements, given that, as we have just seen, such statements can also be expressed by verbs without this morpheme, in an appropriate context.

⁴For a recent study on the use of present tense perfective verbs in Polish generic sentences see Wiemer et al. (2025).

One of the main goals of this chapter is to argue that the morpheme *-va-* (as in (1d)) is best viewed as a generic morpheme *sui generis* (also building on some previous arguments for Czech in Filip (1993b, 1994), Filip & Carlson (1997), Mendia & Filip (2018), Filip (2021), i.a.). This argument implies a rejection of the prevalent traditional view on which this morpheme is a marker of a (possibly unbounded) plurality of situations of the same type (see Genis et al. 2021, and specifically for Czech see e.g., Petr 1986, Dahl 1995, Nübler 2017, and references therein), and as such, it is treated as a marker that falls under imperfective aspect and delimits an (iterative) *Aktionsart* (Poldauf 1964, for instance), or as a kind of pluralization operator deriving a predicate of plural situations. While the notion of a plurality of situations is relevant to the meaning of the morpheme *-va-*, it is neither sufficient nor necessary to ground its meaning, as I will show. Given the erroneous idea that this marker encodes a mere plurality of situations, it is traditionally referred to as the “iterative” marker, and also the “multiplicative”, “frequentative” or “repetitive” marker. That its most common label “iterative” is a misnomer at best, and a category error at worst, is evident in the simple fact that this morpheme is incompatible with iterative adverbials like ‘three times’ or ‘several times’ (see section Section 4). It will also be shown that the marker *-va-* does not have a meaning akin to the adverb of quantification ‘usually’ or the determiner quantifier ‘most’ (*pace* Dahl 1995, i.a.), as in ‘in most situations’.

Related to the main argument to be made here, let me make two clarifications. First, terminology for morphemes and grammatical constructions that are used in sentences that express regularities, patterns, or habits – in short generalizations of various kinds – is not standardized and, as suggested in the previous paragraph, often misleading. Iterativity, multiplicativity, frequentativity, repetitiveness, usuality, pluractionality, and the like are often used interchangeably, and also confounded with habituality and genericity.

When it comes to habituality, it is either viewed as separate from genericity, equated with it (e.g., Comrie 1976), or subsumed under genericity as its special case (e.g., Krifka et al. 1995).⁵

In its widest sense, the term “habitual” is used for all verbs, VPs or sentences that are not episodic. Episodic sentences are true relative to a given world and

⁵For instance, Comrie (1976) uses “habituality” for all the uses of imperfective forms that are not episodic. He divides the semantic domain of imperfectivity into a “habitual” subdomain and an episodic subdomain. However, he confusingly refers to the episodic subdomain as “continuous”, and divides it into two further subdomains: namely, “nonprogressive” and “progressive”. The category “continuous” in (Comrie 1976) is notionally hardly coherent, and formally speaking, there seems to be no language with formal markers for it. For further criticism of Comrie’s classification see Filip & Carlson (1997).

reference time, they do not describe regularities. In this widest sense, habituality has been popularized by Comrie (1976), and is commonly used for one subdomain of imperfectivity, as opposed to its other, namely, episodic subdomain in aspect studies and traditional/descriptive linguistics. Habituality in this sense is, however, problematic, because it subsumes uses of sentences describing what is not generally viewed as a matter of habit at all (Lyons 1977: 716, i.a.), i.e., habit in the sense of regularities of action of typically human, or at least animate, individuals, as in *Fred smokes after dinner*. Some examples are: *Glass breaks easily* (a disposition), *Soap is used to remove dirt* (a function), *Bishops move diagonally* (a rule of a game), *Kim works for the government* (an occupation). Therefore, a better term to cover all types of sentences that are not episodic is *generic*, which is prevalent in formal semantics studies that are informed by philosophy, computer science and psychology.

The Czech marker *-va-*, as in (1d), despite being occasionally labeled “habitual” (e.g., Dahl 1995, Danaher 2003, Karlík 2017), can formally mark not only sentences that describe habits proper (as in *Fred smokes after dinner*), but also generic sentences with kind reference, as in our initial example (1d), and also in other types of generic sentences that describe what is not a matter of habit (see e.g., (50) below).

Therefore, here, for the sake of simplicity, I will use the term *generic* as a cover term for all statements that are *not* episodic, while habituality will be treated as a special case of genericity (as in e.g., Krifka et al. 1995). The term *habitual sentence* will be reserved for a subclass of generic sentences that concern habits proper, namely regularities of action of typically human, or at least animate, individuals (e.g., *Fred smokes after dinner*).

The second point of clarification concerns the confounding of the Slavic generic marker *-va-* (as in (1d)) with the imperfectivizing suffix (as in (1c)). Their confounding has to do with the fact that some of their allomorphic variants seem virtually indistinguishable, and both are used in generic (and habitual) contexts.⁶ While the generic marker *-va-* enforces a generic reading in all its occurrences, the imperfectivizing suffix on secondary imperfectives is compatible with their

⁶Occasionally this confounding may be also motivated and reinforced by speculations about the diachronic development of one of the morphemes from the other, which, however, seem unfounded. For instance, Kuznetsova & Sokolova (2016) assume, without any evidence and only with reference to some assumptions by Kuznecov (1953), that habituality or iterativity is diachronically the original meaning of the Russian secondary imperfectivizing suffix, standardly cited as *-yva/iva-*. According to Kuznecov (1953: 259), it developed as the result of a reanalysis of the the habitual or “iterative” morpheme *-va-* of the verb *byvati* ‘to be usually, typically, normally’ or ‘to tend to be’: *by-va-ti* > *b-yva-ti*. Hence, according to him, the imperfectivizing suffix *-yva/iva-* originally only had the habitual or “iterative” meaning component.

episodic or generic reading. For instance, only the imperfectivizing suffix, but not the generic *-va-*, can be used in sentences with a contextually determined progressive (episodic) reading. I will address the differences between these two morphemes in detail in Section 3, and touch on their differences also elsewhere in this chapter.

With these basic clarifications and caveats out of the way, the main question to be pursued in this chapter is as follows:

QUESTION 1: What properties of the marker *-va-* justify its status as a marker of genericity?

The marker *-va-* constitutes a sufficient, but not a necessary, condition for the expression of generic (and habitual) statements. Such markers are attested in a number of typologically distinct languages. A short list, which also includes Czech, is given in Dahl (1995: 421, fn. 8).⁷ In fact, Dahl uses the Czech marker *-va-* as a paradigm example of this class of markers, which, according to him, fail to qualify as generic markers. Instead, they are to be treated as what he calls “habitual tense-aspect markers” that function as a kind of quantifier over situations with the semantics of ‘most’ or ‘usually’. For Dahl (1995), this claim bolsters his overarching argument concerning the reflexes of the episodic/generic distinction in the grammar of natural languages: namely, if all such verb markers like the Czech *-va-* are not generic markers, but rather habitual markers (‘habitual’ in the sense of Dahl 1995) that fall either under tense or imperfective aspect, then “however sad this may be for those of us to whom the episodic/generic distinction is dear, this distinction more often than not is only indirectly reflected in speakers’ choices between grammatical markers” of aspect or tense (Dahl 1995: 425).

These are the claims that will be contested here. Contrary to Dahl (1995), I will show that the Czech marker *-va-* has properties that preempt its subsumption under tense or imperfective aspect (also *pace* Cable 2022). Its meaning cannot be reduced to a quantifier over situations with the meaning amounting to ‘most’, or ‘usually’, nor any other single expression of quantity or quantification, also contrary to Dahl (1995).

Having established what the marker *-va-* is not, I will show that it derives verbs denoting predicates that exhibit three hallmark properties of generic predicates (see e.g., Carlson 1977, Krifka et al. 1995): namely, (i) they are aspectually

⁷Dahl (1995: 421, fn. 8) gives the following list: Arabic (Classical), Akan, Catalan, Czech, Didinga, German, Guarani, Hungarian, Kammu, Limouzi, Montagnais, Sotho, Spanish, Swedish, Swedish Sign Language, Yucatec Maya, Zulu.

stative, (ii) have a modal (intensional) import which is tied to the quintessentially predictive force of generics, and as such their evaluation requires access to other possible worlds than just the actual one, and (iii) they express generalizations that crucially involve reasoning with exceptions. This then leads me to the next two questions:

QUESTION 2: What kind of generic sentence does the generic marker *-va-* delimit?

QUESTION 3: What is the relation of the generic marker *-va-* to the phonologically null generic GEN operator?

Given that there is no comprehensive semantic analysis of genericity that is commonly accepted, when it comes to the foundational assumptions and terminology in the domain of genericity (and habituality), I will rely on the theoretical background of the quantificational theory of genericity (see e.g., Krifka et al. 1995). Its key hypothesis is that the semantics of generic sentences can be represented by means of a logical (tripartite) structure, where the generic import is captured by means of the phonologically null, modal generic GEN operator, which is akin to overt adverbs of quantification (Q-adverbs) (see Section 2). The quantificational theory of genericity will here serve as a useful point of reference for examining theoretical and empirical issues raised by the generic morpheme *-va-*. It has the advantage that it has a wide interdisciplinary scope, drawing on insights not only from semantics and pragmatics, but also on insights regarding the nature of generalizations, law-like statements and reasoning with exceptions in philosophy, psychology, computer science and other areas of cognitive science.

As I will show, just as GEN, the generic morpheme *-va-* functions as a quantifier over individuals and/or situations, and exhibits variable binding properties that are akin to those of GEN. Moreover, the meaning of both GEN and *-va-* cannot be reduced to any single quantifier, e.g., ‘usually’, ‘most’, or any other quantifier or quantity expression. At the same time, the Slavic generic morpheme differs from GEN in so far as it has two uncancellable inferences: namely, that (i) there are verifying instances in the actual world that count as evidence for the truth of the generalization, and (ii) there are actual or at least potential exceptions or counterexamples to the generalization. The former inference is shared with what are taken to be so-called ‘habitual’ markers in other languages (see e.g., Cable 2022 on the realization inference of Tlingit habitual morphology), and it is also known as the actualization entailment in the genericity literature (see e.g., Green 2000, Hacquard 2006).

The latter inference is clearly manifested in the observation that *-va-* is semantically anomalous or false in generic sentences that have no exceptions in all the possible worlds (such as universal laws of nature), and it must be used in generic sentences that have known counterexamples and would be false without it, all else being equal (see Section 4.1.2). Generally, as I will show, the truth conditions and felicity of generic sentences that are formally marked with *-va-* crucially depend on the speaker's reasoning with exceptions, which reflect the speaker's commitment to the strength of the generalization calibrated in terms of exceptions that (the speaker believes) it has or might have. In effect, generic sentences formally marked with *-va-* are endowed with two layers of modality (intensionality): one that is tied to their generic, law-like, (g)nomonic force, similarly to GEN, and the other to reasoning with exceptions, which may be characterized in terms of the speaker-oriented epistemic and doxastic modality (exploiting general pragmatic principles of interpretation).

The presence of the generic marker *-va-*, and its contrastive absence, in generic sentences differentiates two main types of generic sentences in Slavic languages. Some have argued that different linguistic forms for the encoding of generic statements differentiate different subtypes of generic sentences, each distinguished by different clusterings of formal properties, and possibly requiring different semantic/ontological commitments, and separate semantic/ontological models for their interpretation (see e.g., Greenberg 2003, Pelletier 2009a, Krifka 2012, Boneh & Doron 2013, 2008). Hence, the properties of the Slavic generic marker directly bear on the fundamental question in the domain of genericity (Carlson 1977, 1995):

QUESTION 4: Can we provide a unified semantic analysis for all generic sentences?

A conclusive answer to the above question is still outstanding. A unified analysis for all generic sentences is desirable, and hypothesizing that generic sentences might constitute a single class, markers that are dedicated to the encoding of generic statements, such as the generic marker *-va-*, provide some of the best empirical data for evaluating the claims that address the above question.

The proposal advanced here that the morpheme *-va-* is a generic marker *sui generis*, as well as the empirical and theoretical arguments in its support, have potentially significant consequences for the theory of genericity. First, if the morpheme *-va-* in Czech, in other Slavic languages where it is attested, is a generic marker, as I argue, it would shed at least some doubts on the widespread belief

in linguistics, philosophy and psychology that natural languages have no dedicated markers of genericity (see e.g., Leslie 2008, Johnston & Leslie 2012, Leslie & Gelmann 2012, Rhodes et al. 2018, Cohen 2022, Schiller 2023, to give just a few more recent examples), or only a negligible number (Dahl 1995).⁸ Second, it would support the claims independently made elsewhere (see e.g., Carlson 1977, Schubert & Pelletier 1989, Filip & Carlson 1997, Pelletier & Asher 1997, i.a.), and mostly on semantic grounds, that the episodic/generic distinction is relevant for the grammar of natural language (*pace* e.g., Dahl 1995), and bolster the view on which genericity is a category in its own right, and specifically independent from the tense, modality and aspect (TMA) system.

This chapter is structured as follows. Section 2 introduces the theoretical background of the quantificational theory of genericity. In Section 3, the formation of generic verbs by means of the generic morpheme *-va-*, and its place in the Czech verb system will be outlined in a theory neutral way. Its properties will be also shown not to be aligned with markers of tense or imperfective aspect. Section 4 focuses on its main semantic and pragmatic properties, which together with Section 3 invalidate the grounds for the claim made by Dahl (1995) that the Czech morpheme *-va-* fails to qualify as a generic marker. Section 5 addresses the semantic similarities and differences between the phonologically null generic operator GEN and the Czech generic marker *-va-*. The marker *-va-* introduces a modal (intensional) generic operator VA into the logical representation that differs from GEN in so far as it contributes the uncancellable inference regarding the existence/possibility of exceptions or counterexamples to the generalization which, moreover, must have been realized. The use of the generic *-va-* form, when the corresponding non-generic form might have also been used, gives rise to a rank order (see Horn & Abbott 2012, Horn 2017, and references therein), where the non-generic form is the strong alternative (as it allows no exceptions when used in generic sentences) and the generic form the weak one (given that it requires that there be actual or possible exceptions). The rank order generates either the speaker-oriented certainty or ignorance inference concerning the likelihood of exceptions (a kind of quantity-based implicature), which may be characterized in terms of epistemic and doxastic modality, exploiting general pragmatic principles of interpretation.

⁸For instance, Schiller (2023: 541) observes: “Generics are present in every known language, but in no language are generic sentences ‘marked’ with an explicit generic operator. Generics are studied (by philosophers, linguists, and developmental psychologists) because of their abstruse formal properties and the important role they play in cognitive development.”

2 Genericity: Kind reference and generic sentences

The domain of genericity is divided into two independent, but related, phenomena: namely, KIND REFERENCE and CHARACTERIZING GENERIC SENTENCES (Krifka et al. 1995, Carlson & Pelletier 1995). Some examples are given below:

- (2) a. *Ravens* are widespread. KIND REFERENCE
 b. *Dodos* became extinct in the 17th century.
 c. *Bronze* was invented as early as 3000 B.C.
 d. *The dodo* / **A dodo* became extinct in the 17th century.
- (3) a. *Ravens* are black. CHARACTERIZING GENERIC SENTENCES
 b. *Dogs* bark / *A dog* barks.
 c. Paul plays chess on Saturdays.

There are both formal and semantic reasons for this division. The category of KIND REFERENCE is motivated by the existence of kind-level predicates (KLPs) like *extinct*, *widespread*, *invent* that directly select KIND-DENOTING terms for one of their arguments (in italics in the above examples).⁹ KLPs attribute a property to a whole kind and one which does not hold of its individual members. For instance, in (2b), the property of being extinct holds of the whole kind DODO, and it makes no sense to attribute it to any individual dodo or a closed set of dodos.

The main topic of this chapter concerns CHARACTERIZING GENERIC SENTENCES, or GENERIC SENTENCES for short. Their generic import often arises from the combined meanings of the subject NP/DP and VP, and possibly also in interaction with pragmatic, prosodic (focus), and discourse factors. Generic sentences with kind-denoting terms, as in (3a) and (3b), express a generalization that holds of a kind as well as of its individual members. Generic sentences like (3c) describe a regularity of action ascribed to an ordinary individual, i.e., a habit in the proper sense of this term. While in Krifka et al. (1995) or Carlson (2011), for instance, they constitute a subclass of generic sentences, some take them to be less typical or marginal members of generic sentences (see e.g. Nickel 2017).

All agree that **GENERIC SENTENCES** are fundamentally opposed to **EPISODIC SENTENCES**. The latter refer to particular singular situations, as in (4), or to a closed set of plural of situations, as described in the iterative sentence (5):

⁹A given language may impose specific requirements on the form of a kind-denoting term. In English, for instance, a singular indefinite is often unacceptable.

- (4) Fido is barking at the mailman. EPISODIC sentences
- (5) Paul played three games of chess last Saturday.

However, there is no general agreement on the positive criteria that delimit all and only generic sentences (see e.g., Dahl 1985, 1995, Nickel 2008, 2016, Pelletier 2009a,b, Carlson 2012b). There is some agreement that prototypical examples are “bare” generics like *Birds fly* that involve kind reference in a generic sentence and have no restrictors like Q-adverbs (e.g., *always, generally, usually*), overt determiner quantifiers (e.g., *some, many, most, all*), modal expressions, or qualifiers that restrict the relevant cases based on which the generalization is made.

Moreover, at least since Carlson (1977), there has been some agreement that generic sentences, in contrast to episodic sentences, have three main properties:

- (6) PROPERTIES OF GENERIC SENTENCES
 - a. All are aspectually stative.
 - b. All have an intensional import, which is tied to their predictive force, with implications for other accessible worlds than just the (actual) world of evaluation, and perhaps all possible worlds.
 - c. Most admit exceptions, but may be compatible with no exceptions.

All generic sentences are uniformly stative, be they headed by (i) generic predicates that are derived from stage-level predicates like *bark*, as in *Dogs bark* (on its generic interpretation), or by (ii) individual-level predicates like *(be) intelligent, know (French)*, which are lexically stative predicates (e.g., Chierchia 1995). One standard test for stativity is the lack of reference to particular situations which are locatable at specific spatio-temporal locations. For instance, this is evident in the incompatibility or oddity with adverbials referring to specific time-points and locations:

- (7) a. ?Dogs bark right now.
- b. ?John was intelligent on his porch at 4 p.m.

The reason for the oddity of (7a) and (7b) is that they involve characterizing properties of individuals that hold of them during their whole life-time or at relatively long intervals as well as at any moments of such intervals. Therefore, it is pragmatically odd to assert that they hold only at one specific moment of time or location.

Second, all generic sentences have an intensional (modal) meaning component (first observed by Lawler 1973, Dahl 1975), which follows from the nature of generalizations. They transcend our immediate experiences of the world, particular

situations, states of affairs, or facts, in so far as they specify what is possible, likely, i.e., they have a predictive, law-like force. This may be based on what actually obtains at given worlds and times, and we might have observed. However, some generics only concern what is *merely* possible and may even never ever be realized, as in (8a) and (8b). They provide the most compelling evidence for intensionality of generics:

- (8) a. The Speaker of the House succeeds the Vice President. (Carlson 1995)
(Context: The rule of the United States Constitution that has never been enacted.)
- b. This knife cuts cheese.
(Context: The knife has just been delivered in the mail, but its blade got irreparably damaged in shipping.)

The intensional component of generic sentences may be grounded in some underlying causes, principles or factors, or some kind of law that we may surmise behind the pattern, which we may know, but also about which we may be agnostic or ignorant. For instance, *Dogs bark* is true based not just on inductive evidence (e.g., having witnessed barking dogs on a number of occasions), but also on evidence which may not be directly accessible to us, like some underlying causes in the world (such as regularities governing the biology of living organisms) which ultimately ground observable behavior like dog barking.

The third key property of generic sentences is their exception-tolerance. There are generics that admit of no exceptions in all the known conditions, e.g., universal laws of nature like *The Earth revolves around the Sun* or *Water consists of two hydrogen atoms and one oxygen atom*. However, most tolerate exceptions, but may also be compatible with no exceptions whatsoever. For instance, we judge *Dogs bark* to be true, because most dogs bark, even if, as is generally known, there are non-barking dog species like Basenjis and Greyhounds, or individual dogs that do not bark due to injury or illness. These are exceptions that we tolerate. As an extensional consequence, it follows that we cannot validly infer for any particular, arbitrary individual dog that it will have the property of barking. Put differently, that dogs bark is a defeasible fact about dogs. This makes generic sentences particularly amenable to analyses in terms of non-monotonic logics.

What makes any analysis of exception-tolerance so intriguing and challenging is that different types of generic sentences admit a different number and kind of exceptions or non-confirming cases, while still remaining true. Some we judge to be true based on majority satisfaction, such as *Dogs bark*. However, majority satisfaction is neither sufficient nor necessary for the truth of all exception-tolerating generics. It is not sufficient, because, for instance, (9a) is false, despite

the majority of printed books being paperbacks. Majority satisfaction is not necessary, because what is characteristic of a kind need not be prevalent among its members. (9b) holds of only a minority subkind, male adult lions. What is characteristic of a kind may also be characteristic of only a minute minority, as (9c) shows. It is judged true, even if a vast majority of mosquitoes (99 %) fail to carry the virus.

- | | | |
|-----|---|-------|
| (9) | a. Printed books are paperbacks. | FALSE |
| | b. Lions have a mane. | TRUE |
| | c. Mosquitoes carry the West Nile virus (though 99 % do not). | TRUE |

How many and what kind of exceptions a given generic sentence will admit depends on a variety of factors which may include not only objective facts of the world, but also our kind-specific expectations, which may be influenced by our subjective judgments about what we evaluate as significant, salient or striking in some way for that kind, like a lion's mane or a virus that is highly dangerous to us. Compatibility of generics with a different number and kind of exceptions raise the fundamental question about what we view as "characteristic", "typical" or "normal", on the one hand, and what we view as failing to conform to a given generic statement and why, on the other hand.

The above examples and observations should suffice to illustrate that intensionality and exception-tolerance foil any attempt to analyze all generic sentences in terms of a *single* extensional quantifier (over episodic formulas containing free variables) (e.g., *all*, *most* or *some*), a *single* quantity expression (e.g., *in a majority of cases* or *in a significant number of cases*), quantity-based criterion, or statistical criterion/correlation, no matter how contextually constrained and/or modalized they might be, or possibly also defined in vague or probabilistic terms (Pelletier & Asher 1997, Carlson 2012b, Nickel 2013, 2017, i.a.).

A number of promising analyses have been proposed for exception-tolerance of generics and related issues, while intensionality has garnered much less attention (see summaries in Krifka et al. 1995, Pelletier & Asher 1997, Mari et al. 2013, Nickel 2017, i.a.). None of the proposed analyses has offered a satisfactory comprehensive semantic framework for all generic sentences, and none so far has emerged as widely accepted.

Therefore, when it comes to representational assumptions, as a useful point of reference and an organizing principle for examining meanings of generic (and habitual) sentences here, I will at various points rely on a logical representation in terms of a tripartite structure with a phonologically null, modalized dyadic operator GEN, akin to overt Q-adverbs, used in quantificational theory of genericity.

In its original formulation, it has the following general form (adapted from Krifka et al. 1995: 32, ex. (57), which in turn relies on suggestions in Krifka 1987):¹⁰

- (10) $\text{GEN}[x_1 \dots x_i; y_1 \dots y_j](\text{RESTRICTOR}[x_1 \dots x_i]; \text{MATRIX}[\{x_i\}, y_1 \dots y_j])$
 $x_1 \dots x_i$: variables bound by GEN
 $y_1 \dots y_j$: variables bound existentially, with scope just in matrix
 $\{x_i\} \dots \{x_i\}$: means $x_1 \dots x_i$ may or may not occur in matrix

In the tripartite structure (10), just like an overt Q-adverb, GEN relates two arguments, a Restrictor and a Matrix which are separated by ‘;’. GEN stands for a certain characterizing relation between the conditions stated in the Restrictor (which states the restricting cases) and Matrix (which makes the main assertion of the sentence): intuitively, it sets conditions on the number or proportion of individuals that may satisfy the formula, and if the conditions stated in the Restrictor hold, other conditions stated in the Matrix will also hold.

We may distinguish three main types of instances or particulars that can serve as the base for the generalization (Carlson 2011): (i) instances of particular INDIVIDUALS (as in *Ravens are black*), (ii) instances of particular SITUATIONS involving (stages of) a single individual of an ordinary sort (as in *Paul plays chess*), or (iii) instances of particular SITUATIONS involving instances of different INDIVIDUALS that are members of a given kind (as in *Dogs bark*).^{11,12}

The cases (ii) and (iii) involve quantification over situations, which in Krifka et al. (1995) is a defining feature of what they refer to as HABITUAL SENTENCES, a subcase of GENERIC SENTENCES:

- (11) A sentence is HABITUAL iff its semantic representation is of the form
 $\text{GEN}[\dots s \dots; \dots](\text{RESTRICTOR}[\dots s \dots]; \text{MATRIX}[\dots s \dots]),$
 where s is a situation variable (Krifka et al. 1995: 32, ex. (56))

¹⁰GEN and overt Q-adverbs are not unselective binders, in contrast to their early treatments, as in Carlson & Pelletier (1995), for instance. Which variables can be bound in a given logical structure depends on a number of factors, including context and topic-focus structure.

¹¹The notion of SITUATIONS corresponds to occasions or cases, as Lawler (1972) originally proposed, and are akin to “cases” of Lewis (1975) (Krifka et al. 1995: 30). It plays a role similar to that of STAGES in Carlson (1977, and elsewhere) which is tied to Carlson’s distinction between stage-level (episodic) predicates (SLPs) and individual-level (stative) predicates (ILPs).

¹²Generics that fall under (iii) involve a “double generalization” (in Carlson 2011; see also Pelletier & Asher 1997: 1131): *Dogs bark* is true by virtue of (a) *individual dogs* that realize the kind DOG having the characterizing property of barking, and also by virtue of (b) particular *situations* of barking by a stage of an individual dog.

Here, I reserve the term *habitual sentence* for only generic sentences that express habits in the proper sense of the word, as also observed above (Section 1). Most importantly, and as we will see below in Section 4, the generic morpheme *-va-* is not constrained to conveying habits proper, i.e., regularities of action of animate, mostly, human agents (the case (ii) above). Neither does it require that its domain contains situations (the cases (ii) and (iii) above), but it may quantify over individuals only (the case (i) above). In short, it functions as a quantifier over all the three main types of particulars mentioned above, and in this respect it is akin to GEN.

For any theory of genericity the main burden of the analysis is on specifying the relation between the generalization and what it is based on, its basis for generalization, which in many cases (but not all) are some particulars, instances, and particular facts. How this relation comes about is at the core of what it means for a sentence to express a generalization, rule, pattern and the like. We have a rich variety of competing proposals regarding this relation, and by the same token regarding the truth conditions of generic sentences that underpin GEN. One useful way of sorting them out was proposed by Carlson (1995), who suggests that they can be divided into two main perspectives, and mixed proposals combining some elements from both can also be found. One perspective is characterized in terms of an *INDUCTIVE MODEL*, on which generic sentences express inductive generalizations that are based on some observed (or unobserved) set of instances or particulars in the actual world from which we infer a general rule. Paradigm cases are sentences like *Paul plays chess* that express generalizations over situations of the kind described by their corresponding episodic sentences like *Paul was playing chess when I called him yesterday*, *Today during the lunch break, Paul played chess with Erwin*, *Last week, Paul played chess with grandpa and won each and every time*. On this perspective, generics would (mainly) express descriptive (weak) generalizations.

The second main perspective corresponds to what Carlson (1995) dubs a *RULES-AND-REGULATIONS MODEL*. On this perspective, generics express generalizations that do not (necessarily) depend on particular episodic conditions in the actual world, but rather on some causal structure or underlying forces in the world, which give rise to episodic instances (if there are any), and motivate the patterns that we recognize in the actual world. Paradigm cases are generic sentences expressing various rules, such as game rules that we may learn directly, rather than inferring them from episodic conditions in the actual world, for instance, by reading the relevant rules, manuals or guidelines, such as the rules of chess like *Bishops move diagonally*, or following instructions, customs and the like.

In contrast to the inductive model, the rules-and-regulations model can also handle generic sentences that are independent of particular instances or situations, because they have no actualized verifying instances, such as (8a). On the rules-and-regulations model of genericity, we judge generic sentences to be true or false with respect to a set of rules (or a finite list of propositions), which are considered to be irreducible entities, supporting the truth of possibly an unlimited number of propositions.

Carlson (1995) proposes that the rules-and-regulations model is a better candidate than the inductive one for a unified semantic analysis of generic sentences, and a model which might ground the semantics of GEN. However, it is not a good fit for generics like *Paul plays chess*, “which lend the most prima facie plausibility to the inductive model” (Carlson 1995: 237). A rules-and-regulations model requires us to identify some rule, causal factor or underlying condition that gives rise to or “regulates” Paul’s chess playing behavior, which seems rather far-fetched.

These two main perspectives raise an intriguing question which of them might best fit generics marked with the generic morpheme *-va-* in Czech, and in generic sentences that are marked with the cognate marker in other Slavic languages. Given that the marker *-va-* requires that there be realized instantiations that count as evidence for the truth of generic sentences marked by it, as I will show, generic sentences marked with *-va-* at first blush fit the inductive model of genericity. At the same time, as it will be shown, *-va-* can also be used in generics that express rules, regulations, often with the goal of ensuring plausible deniability, or safeguarding a rule, a regulation against refutation by states of affairs of which its author is ignorant. The exploration of the properties of the generic morpheme *-va-* is highly relevant to the answer to the questions posed at the outset: What kind of generic sentence does the generic morpheme *-va-* delimit? Can we provide a uniform analysis for all generic sentences?

3 Morphological background

3.1 The perfective/imperfective and the generic/non-generic distinctions

The generic marker *-va-*, here glossed as GEN, is productive in West Slavic languages: namely, in Czech, Slovak (e.g., Kučera 1981, Bílý 1986, i.a.), and to a lesser degree also in Polish (Bílý 1986, i.a.). In South and East Slavic languages, its oc-

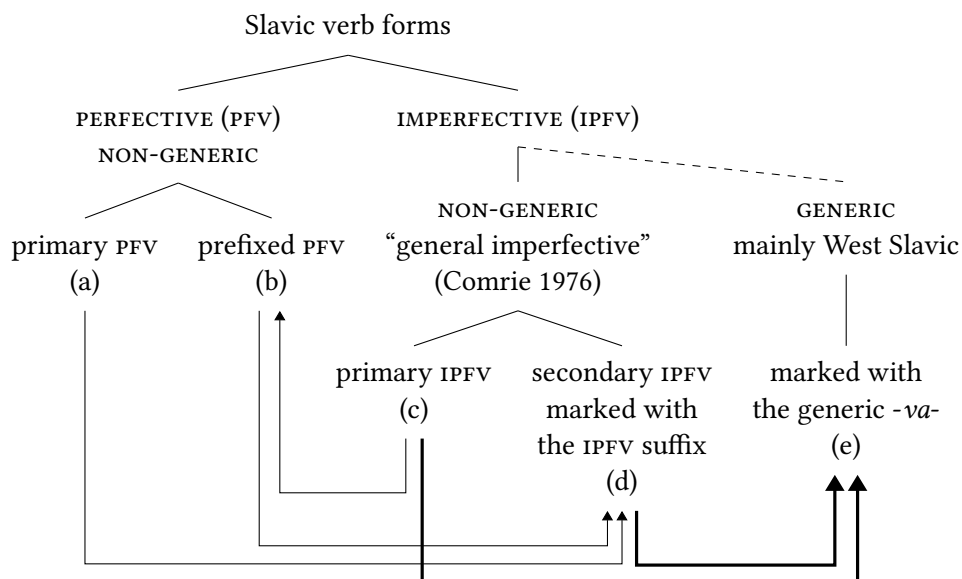


Figure 1: Common formation patterns of non-generic verb forms (perfective and imperfective) and generic verb forms (following e.g. Petr 1986: 180). The dashed line: The traditional subsumption of generic (also known as “iterative”) verbs under imperfective aspect is not uncontroversial (see e.g. Kopečný 1948, i.a.) and may best be refuted.

currences are also attested, but they may be treated as a part of a lexicalized combination with a verb base.¹³

Taking Czech as our case in point, in traditional reference grammars, the verb system is structured by two main distinctions: (i) the perfective/imperfective distinction, and (ii) the generic/non-generic distinction. This is sketched in Figure 1, which builds on the rudimentary schema given in Petr (1986: 180). Table 1 contains examples illustrating the different types of verb forms and their morphological relations.

In Bohemistics, the morpheme *-va-* (here glossed as GEN) is traditionally labeled “iterative” (*iterativní* in Czech) or “multiplicative” (*násobená* in Czech). It

¹³In Russian, verbs with this marker are taken to be a feature of a colloquial (“substandard”) speech (Isačenko 1962: 405–407, Forsyth 1970: 28 and 168–171, Comrie 1976: 27, Kučera 1981: 177, Bílý 1985, Vinogradov 1986: 413–414, Petr 1986, Bílý 1986), while some of them are in fact quite frequent: e.g., *byvat’* (< *byt’* ‘to be’), *znavat’* (< *znat’* ‘to know’), *govarivat’* (< *govorit’* ‘to speak’), *pivat’* (< *pit’* ‘to drink’), *siživat’* (< *sidet’* ‘to sit’) and *xaživat’* (< *xodit’* ‘to go’). Interestingly, the generic marker similar to that in West Slavic languages is productive in some Northern Russian dialects (Barnetová 1956).

is clearly separated from the imperfectivizing suffix (see e.g., Petr 1986, Lamprecht et al. 1986, Kosek 2014, Biskup 2024; Genis et al. 2021 also separate the cognate generic morpheme from the imperfectivizing suffix in other Slavic languages). However, at the same time, this morpheme is subsumed under imperfective aspect, treated as a kind of imperfective marker, with one notable exception, namely Kopečný (1948, 1966).

Notice that Figure 1 has a dashed line connecting the IMPERFECTIVE (IPFV) with the GENERIC node. It indicates that the traditional subsumption of verbs with generic morpheme *-va-* under imperfective aspect is not uncontroversial. It is one of the main goals of this chapter to show that verbs marked with the generic morpheme *-va-* do not neatly fit the imperfective paradigm in Slavic languages, and imperfectivity in natural language generally. They have a number of semantic and pragmatic properties that differ from both primary and secondary imperfectives in Slavic languages. This in turn conforms with independent arguments made elsewhere that genericity is best viewed as independent of imperfectivity, and other categories of the TMA system (see e.g., Filip & Carlson 1997).

In Slavic languages, all the forms without the generic morpheme *-va-* that are inherently episodic alternate between episodic and generic readings depending on context. Hence, they are referred to as non-generic in Figure 1. Slavic non-generic forms are either imperfective or perfective, and members of both classes can be either simple (root, primary) or derived, mostly by means of derivational affixation. Primary and secondary imperfectives correspond to “general imperfectives” (Comrie 1976), meaning that they cover the whole semantic domain of imperfectivity, both its episodic and generic subdomains.

Perfective verbs in Slavic languages, all of which are non-generic (i.e., have either episodic or generic uses depending on context), are for the most part derived. Prefixation operating on primary imperfective bases is the most common way of forming perfective verbs. This corresponds to the derivational relation (c) → (b) in Figure 1. Slavic verb prefixes have hallmark properties of lexical-derivational, rather than inflectional, morphology, and hence are not grammatical markers of perfectivity akin to bona fide inflectional morphology like tense markers (*pace* Borer 2005, i.a., for arguments see Filip 1992, 2000, 2026, and elsewhere).¹⁴ Slavic prefixation derives new verbs, typically with a different lexical meaning and argument structure than their derivational base. In the relevant examples in Table 1,

¹⁴The argument that Slavic verb prefixes are not grammatical markers of perfective aspect in Slavic languages provides a key support for the view that the Slavic perfective/imperfective distinction is predominantly of lexical-derivational, rather than inflectional, nature. This is also reflected in the common Slavic lexicographic practice of marking lexical entries of verbs as either perfective or imperfective, and some as bi-aspectual.

Table 1: Examples illustrating the morphological relations included in the schema in Figure 1

(a) dát ^{PFV} primary give.INF 'to give'	→	(d) dávat ^{IPFV} secondary give.IPFV.INF 'to give'/'to be giving'
(d) dávat ^{IPFV} secondary give.IPFV.INF 'to give'/'to be giving'	→	(e) dávávat ^{generic} give.IPFV.GEN.INF 'to give regularly, seldom, often, usually, ...
(c) štěkat ^{IPFV} primary bark.INF 'to (be) bark(ing)'	→	(b) poštěkat ^{PFV} prefixed PREF.bark.INF 'to bark (a little)'; metaphorically: 'to bicker', 'to squabble'
(b) poštěkat ^{PFV} prefixed PREF.bark.INF 'to bark (a little)'; metaphorically: 'to bicker', ...	→	(d) poštěkávat ^{IPFV} secondary PREF.bark.IPFV.INF 'to (be) bark(ing) (a little)'; metaph.: 'to (be) bicker(ing)', ...
(d) poštěkávat ^{IPFV} secondary PREF.bark.IPFV.INF 'to (be) bark(ing) (a little)'; 'to (be) bicker(ing)', ...	→	(e) poštěkávávat ^{generic} bark.IPFV.GEN.INF 'to (tend to) bark/bicker a little, regularly, seldom, often, usually, ...
(c) štěkat ^{IPFV} primary bark.INF 'to (be) bark(ing)'	→	(e) štěkávat ^{generic} bark.GEN.INF 'to (tend to) bark regularly, seldom, often, usually, ...

the prefix *po-* not only derives a perfective verb from the primary imperfective base, under (c) in Figure 1, but also contributes an attenuative meaning to the resulting prefixed verb which roughly amounts to ‘bark (a little)’, and this prefixed verb also has a metaphoric sense of ‘to bicker’, ‘to squabble’. Prefixes are also applied to already prefixed perfective bases, so that more than one prefix can occur on one and the same verb.

Secondary imperfectives (under (d) in Figure 1) are derived by means of the imperfectivizing suffix, glossed as IPFV. In all Slavic languages, the imperfectivizing suffix applies to perfective bases and derives their imperfective counterparts, typically with the same lexical meaning and argument structure. The perfective bases to which the imperfectivizing suffix is applied is either primary perfective, as in (a) → (d), or prefixed, as in (b) → (d).¹⁵

Most importantly, in contrast to the Slavic imperfectivizing suffix, the generic morpheme applies to an imperfective base realized in verb forms that can have either an episodic or a generic reading and derives a form that is unambiguously generic, i.e., it excludes any episodic interpretation.¹⁶ The imperfective base to which the generic morpheme *-va-* applies is either primary, as in (c) → (e), or secondary, as in (d) → (e). When the generic morpheme is applied to a secondary imperfective base, as in (d) → (e), the resultant generic verb contains both the imperfectivizing suffix (IPFV) and the generic morpheme (GEN).

In Czech, the derivation of specifically generic forms from imperfective non-generic ones is transparent, regular and productive, and verbs with the generic morpheme *-va-* are common in all styles of speech (Kučera 1981: 177, Petr 1986).¹⁷ The Czech generic morpheme *-va-* can be reduplicated, which also confirms its productivity: e.g., *mít*^{IPFV} ‘to have’ > *mí-vá~vá~va-t* [have-GEN~GEN~GEN-INF] ‘to tend to have, very sporadically/very often’, ‘to have once in a great while’. Attested reduplication examples are fairly easy to find, and two are given below:

- (12) a. *Říkávává se, že po bitvě je každý generál.*
 say.GEN.GEN.3SG.PRS REFL that after battle is everybody general
 ‘As the saying goes, after the battle, everybody is a general.’

¹⁵The imperfectivizing suffix may generate so-called “aspectual pairs”, i.e., two verbs with the same lexical meaning that only differ in grammatical perfective/imperfective aspect. Therefore, some classify it as an inflectional suffix (Isačenko 1960, Filip 1993a, 1999). It is also often taken to be the only marker in Slavic languages that is solely dedicated to the encoding of aspect (Isačenko 1960, Filip 1993a, 1999).

¹⁶In Czech, the generic morpheme *-va-* applies to past tense imperfective stems (Petr 1986: 184).

¹⁷There are also a few irregular generic verbs: e.g., in Czech *číst* > *čítat* ‘to read’, *vidět* > *vidat* ‘to see’, *slyšet* > *slýchat* ‘to hear’, *ležet* > *léhat* ‘to lie (down)’.

- b. Bude to z jiného konce, než býváva zvykem.
 will.be it from different end than be.GEN.GEN.3SG.PRS customary
 ‘It will be approached from a different angle than is customary.’

Reduplication is often associated with the speaker’s subjective judgment concerning the regularity of instances that form the base for the generalization. It may affectively connote their strikingly high or low frequency.¹⁸

The formation of generic verbs by means of the morpheme *-va-*, as just illustrated with some Czech examples, follows a cross-linguistic pattern that is attested in a number of typologically unrelated languages that have specifically generic forms: namely, generic forms are derived from non-generic ones; they are the marked case relative to non-generic forms, which are morphologically or syntactically simpler and alternate between episodic and generic interpretations. We do not find generic forms that are basic and episodic forms derived from them (Carlson 1995, 2005, and elsewhere).

Examples of languages that have verb markers enforcing only a generic reading of sentences, but are not a necessary condition for the expression of generic statements, can be found in Dahl (1985, 1995). The Czech marker *-va-* serves as his example par excellence for this class of markers. However, Dahl (1995) also argues that they are not generic markers, but rather habitual markers falling either under tense and/or aspect; they are habitual markers, because, according to him, a bona fide generic marker must constitute not only a sufficient but also a necessary condition on the expression of all and only generic statements. Only 3 languages in Dahl’s data base of 76 languages have such a generic marker: namely, Wolof (Niger-Kordofanian), Işekiri (Niger-Kordofanian), and Maori (Austronesian) (Dahl 1995: 420, 424). Such observations constitute one of Dahl’s empirical arguments in support of his main conclusion that the episodic/generic distinction has no direct relevance for the grammar of natural language; rather, it is generally only indirectly reflected in speakers’ choices between grammatical markers (Dahl 1995: 425).

To this it might be immediately objected that Dahl’s claim here relies on an overly restrictive, unwarranted requirement of what it means for a given marker to count as a grammatical marker of a given category. As is well-known, bona fide grammatical markers (e.g., the English past tense morpheme *-ed*) do not neatly delimit a single semantic domain, and absolute obligatoriness is not a necessary criterion for grammaticalization, as “[s]omething is obligatory relative to the context; i.e., it may be obligatory in one context, optional in another and

¹⁸(12a) retrieved at <https://www.vidlakovykydy.cz/clanky/magnotova-linie-ceskoslovenske-opvevneni> accessed 13/12/2021; (12b) from a native Czech speaker (p.c.).

impossible in a third context” (Lehmann 1995: 12; see also Heine & Kuteva 2002, 2007, and others, for similar observations).

Dahl (1995: 421) also observes that “[a] relatively safe generalization is that habituals in general exhibit a low degree of grammaticalization. They are [...] possible sources of a grammaticalization path yielding what would be the primary candidates for a gram that systematically marks genericity.” In West Slavic languages at least, and as we have just seen in Czech, the generic marker is productive, and systematically in all its occurrences only contributes a generic interpretation to the meaning of a verb, in a completely transparent and compositional way. This would seem to indicate that it exhibits a high degree of grammaticalization.

Having sketched some basic morphological facts, the next section will address the independence of the Slavic generic morpheme from tense and imperfective aspect, and specifically its independence from the imperfectivizing suffix. We will also see that it does not neatly fit into the Slavic imperfective paradigm, contrary to standard assumptions in Bohemistics, and also contrary to Dahl (1995).

3.2 Independence of the generic morpheme from tense and imperfective aspect

The generic morpheme *-va-* is independent of tense. It occurs on non-finite forms (including the infinitive and imperative, the relevant examples are given in Filip & Carlson 1997), and with all the overt tense markers (past, present, and future) on the same verb or in the same verb complex, as we see in the following Czech examples:¹⁹

- | | | | |
|------|----|--|---------------|
| (13) | a. | Karel hrával hokej. | PAST TENSE |
| | | Charles play.GEN.PST hockey | |
| | | ‘Charles used to play hockey.’ [remote past] | |
| | b. | Karel hrává hokej. | PRESENT TENSE |
| | | Charles play.GEN.PRS hockey | |
| | | ‘Charles tends to play hockey.’ [i.e., regularly, often, seldom, ...] | |

¹⁹In the past tense, generic sentences formally marked with *-va-* have a remote past time reference, whereby its reduplication entails a high degree in the past temporal remoteness (Kučera 1981, 1983, Bílý 1986): e.g., *Vídával* [see.GEN.GEN.1SG.PST] *jsem ho tam ve čtvrtky* ‘I used to see him there on Thursdays once in a while / regularly ... a very long time ago.’ (Chmelíčková 2005). Generally, *-va-* (possibly also reduplicated) combined with the past tense triggers a conversational implicature that the described situation no longer holds, and can also refer to single extended situations in the past.

- c. Karel bude hrávat hokej. FUTURE TENSE
 Charles be.AUX.FUT play.GEN.INF hockey
 'Charles will tend to play hockey.' [i.e., regularly, often, seldom, ...]

The independence of the generic morpheme *-va-* from tense is not surprising, given that generally markers of genericity and tense are orthogonal to one another in languages that have overt markers for genericity and tense. There are tensed languages with generic markers (e.g., Czech, Polish, Slovak), without generic markers (e.g., English), and tenseless languages with arguably generic markers (e.g., American Sign Language) and without them (e.g., Chinese), in which case they convey generic statements by other means (e.g., in Dyirbal and Burmese by means of a modal distinction between realis and irrealis; see Comrie 1985: 51).

Also notionally, genericity (and habituality) and tense are independent of each other. Tense is a deictic category, while genericity (and habituality) are non-deictic. In tensed languages like English, generic sentences can be in any tense form, and refer to present, future or past time: e.g., *Dinosaurs ate kelp*; *Mary walks to school along the river*; *When he retires, he will spend more time with his family*. In English, the simple (non-progressive) present tense, as opposed to the present progressive, preferably has a generic interpretation.

Generic sentences are compatible with any sufficiently large temporal interval, contextually located in the present, past, or future. They may also express generalizations that contain overt or covert spatial/temporal restrictors related to the present, past, or future, as shown in (14).

- (14) a. *Současný prezident jídvává hamburgery.*
 current president eat.GEN.PRS hamburgers
 'The current president eats hamburgers.'
- b. *Dinosauri jídváli kelp.*
 dinosaurs eat.GEN.PST kelp
 'Dinosaurs (usually) ate kelp.' [The generalization is restricted to the distant past by virtue of the general world knowledge about the dinosaur kind.]
- c. *Od pondělka se bude tato kancelář otvírávat až ve 2 hodiny.*
 from Monday REFL be.AUX.FUT this office open.GEN.IPFV only at 2 o'clock
 'Starting next Monday this office will open only at 2 p.m.'

Let me now turn to the independence of the generic marker from the imperfectivizing suffix. It is not uncommon for the generic marker *-va-* and the imperfectivizing suffix to be confounded, both notionally and formally, given that their surface allomorphic variants may often seem alike, and both occur in generic sentences. One compelling piece of evidence for their formal independence stems from the fact that they may co-occur on the same verb (see Filip & Carlson 1997). Consider the following attested example:²⁰

- (15) Bývalý mi taky *dávával* nádherně a originálně
 ex me.DAT also give.IPFV.GEN.PST wonderfully and originally
 vázané kytice, přímo umělecká díla, ...
 bound bouquets directly art works
 ‘My ex also used to give me wonderful bouquets in original
 arrangements, real works of art, ...’

In (15), the morpheme sequence which is here glossed with IPFV.GEN is the result of the derivation pattern (a) > (d) > (e), given in Figure 1. In the first step (a) > (d) the imperfectivizing suffix (IPFV) applies to a (primary, or root) perfective base and derives an imperfective form, which alternates between an episodic and a generic reading depending on context. Hence, it is unmarked for genericity, i.e., non-generic or a “general imperfective” form (in the sense of Comrie 1976). In the second step (d) > (e), the generic *-va-* ‘GEN’ is added “on top of” the imperfectivizing suffix (IPFV), and yields a verb that only has a generic reading.

Following the standard assumption about what it means to be a “formal member of a category system”, given that the Slavic imperfectivizing suffix (IPFV) is a marker of imperfectivity, as all agree, the morpheme *-va-* ‘GEN’ cannot be a different marker of the same imperfective category. A given formal member of a category system stands in a complementary distribution to other formal members of the same category, morphologically and syntactically. For instance, the present and past tense morphemes in English are formal members of the same tense category, hence are in complementary distribution: e.g., **she laugh-s-ed* or **she laugh-ed-s*. The markers of tense and progressive aspect in English can co-occur, because they are *not* formal members of the same grammatical category: e.g., *she was/is/will be laughing*. If we accept this line of reasoning for the present and past tense morphemes in English, we should be ready to accept that the fact that the generic marker *-va-* and the imperfectivizing suffix do not stand in a

²⁰The example is from <https://abecedazdravi.cz/diskuse/psychicke-problemy-a-psychologie/deprese-i-nova-diskuze/89>; accessed on July 9, 2023.

paradigmatic opposition to each other indicates that they are not only two separate morphemes, but also that they cannot be two different markers of the same grammatical category system, and specifically both marking imperfectivity.

To the above argument it could be objected that the morpheme sequence which is here glossed with IPFV.GEN (as in (15)) instantiates two occurrences of one and the same imperfectivizing suffix: ‘IPFV + IPFV’. Hence, what we have in (15) is something like a doubly aspectualized imperfective verb associated with a generic (or habitual) reading. Implicit behind this idea would seem to be some encoding economy principle, namely that it is superfluous to apply twice one and the same imperfectivizing suffix with one and the same imperfective function. Therefore, the second application of IPFV must have a different function, i.e., affect a truth-conditional change from a non-generic, or general imperfective, to a generic imperfective meaning.

One immediate *prima facie* counterargument against the ‘IPFV + IPFV’ double aspectualization idea has to do with verbs that only have a generic interpretation, but contain a *single* occurrence of the overt alleged IPFV suffix. This is a subclass of generic verbs derived following the pattern (c) > (e) in Figure 1. In order to rescue the double aspectualization idea it might be proposed that simplex (primary) imperfective verbs like *štěkat* ‘to (be) bark(ing)’, in Figure 1 under (c), contain a phonologically null imperfectivizing suffix $\emptyset^{\text{IPFV}}$, and so generic verbs that are formed following the pattern (c) > (e) contain one covert and one overt instance of the same IPFV suffix: $\emptyset^{\text{IPFV}}$ + IPFV.²¹ However, such a move is hardly empirically motivated at all, theoretically costly, and generally supports a proliferation of empty categories for purely theory-internal reasons.

Reduplication without a change in the truth conditions is common in natural languages, it may serve to add affective connotations, intensification, as we have seen above in (12a) and (12b). We would have to ask: Just why should the reduplication of two overt IPFV morphemes have a truth-conditional consequence, and enforce the generic interpretation, when a truth-conditional change is not a necessary consequence of all reduplication processes? Similarly, we may ask: Just why should a double aspectualization consisting of one overt IPFV morpheme and one covert $\emptyset^{\text{IPFV}}$ morpheme have a truth-conditional consequence such that the second overt occurrence of this morpheme restricts the verb to only a generic interpretation?

Moreover, the IPFV morpheme occurs on verbs that are general imperfective verbs, because they can alternate between an episodic (including progressive)

²¹That Slavic primary imperfectives introduce a null imperfective suffix is proposed in syntactic accounts of Slavic aspect, see e.g. Grønn (2026 [this volume]).

and a generic interpretation depending on context (as imperfective morphemes in natural language do, see e.g., Comrie 1976). Presumably the second overt occurrence of the alleged IPFV should be also inherently amenable to either the episodic or generic interpretation, but then it is unclear just why the second overt occurrence of IPFV should enforce the generic interpretation, and not the episodic interpretation of its output. For instance, it is unclear why it does not constrain its output to only the episodic progressive interpretation. The specifically generic properties of verbs containing the sequence IPFV+IPFV or the sequence $\emptyset^{\text{IPFV}}$ +IPFV would have to be postulated, because they do not follow from the semantics of either IPFV or $\emptyset^{\text{IPFV}}$.

A more sophisticated version of the reduplication or double aspectualization idea is proposed by Cable (2022): namely, the Czech “habitual” (in his terms, generic in my terms) morpheme, as explored in Filip (2018), is, according to Cable “etymologically an instance of imperfective-aspect combining with another, lower imperfective-form of the verb, a kind of doubly aspectualized verb (Filip 2018). One might wonder, then, whether that higher imperfective-morphology might at all synchronically be a realization of a (bound) T-head, while the lower aspectual morphology is a realization of (non-modal) [IMPRV_{OG}]” (Cable 2022: 44–45) [where OG stands for “ongoing”, HF]. This suggestion builds on Cable’s analysis of habitual morphology in Tlingit (Na-Dene; Alaska, British Columbia, Yukon), where a verb may be realized in the “habitual perfective-mode” formed with the habitual suffix *-ch*, or in the “habitual imperfective-mode” consisting of the imperfective form followed by the habitual particle *nooch*. The main thesis is that there are two different morphosyntactic structures that underpin habituality both within and across languages. One where the habitual semantics is directly contributed by imperfective aspect, which has the meaning of a modal operator quantifying over alternative worlds/situations (see e.g., Ferreira 2016). In Tlingit, it is only verbs bearing imperfective-mode that can be used to describe pure dispositions, non-actualized regularities. The other structure underpinning habituality corresponds to the bi-partite morphosyntactic structure, where the habitual morphology is the realization of a quantificationally bound T(emporal)-head, licensed when it is bound by a (possibly covert) Q-adverb, which combines with the lower perfective or imperfective Asp-head to yield a complex form with a habitual meaning.²² The Tlingit habitual morphology is a kind of quantificationally dependent tense, where it is the (potentially covert) Q-adverb that is

²²This licensing strategy resembles that of Chierchia (1995), who assumes that genericity is subsumed under aspect, and “all languages have a distinctive habitual morpheme (say, *Hab*) which can take diverse overt realizations” (Chierchia 1995: 197). Chierchia’s *Hab* is a functional head in an (imperfective) aspectual projection, and either realized by explicit aspectual morphemes,

the source of the habitual meaning. It is analyzed as a temporal quantifier, e.g., ‘always’, ‘sometimes’, which quantifies strictly over times in the actual world, it does not introduce modal quantification over other possible worlds, because, according to Cable, it comes with the actualization entailment.

Cable’s (2022) double aspectualization proposal for Czech does not work for three main reasons. First, it raises the questions posed above about just why the verb “doubly aspectualized” with the same imperfectivizing suffix should have its interpretation constrained to the generic (Cable’s “habitual”) reading, when the imperfectivizing suffix covers the whole imperfective domain, both its episodic and generic subdomains. Why does not the “double imperfective aspectualization” exclude the generic interpretation, and enforces only an episodic (e.g., progressive) interpretation?

Second, Cable (2022) proposes that habitual morphology, within and across languages, is a kind of quantificationally dependent tense. However, generally, the semantics of generic sentences and habitual sentences as their special case cannot be analyzed in terms of temporal notions, or reduced to the semantics of Q-Adverbs (see Section 2, and references therein). As has been shown above, genericity, and habituality as its special case, is independent from tense, both notionally and formally. Specifically, the Czech generic morpheme *-va-* (a habitual morpheme in terms of Cable 2022) is fundamentally independent of tense.

Third, as many others, Cable (2022) correlates the so-called actualization entailment of habitual sentences with the lack of intensionality (modality). However, all generic sentences, including habitual sentences (i.e., concerning habits in

or, as in English, or it is covert. *Hab* carries the agreement feature [+Q] which in its specifier requires some overt Q-adverb (Chierchia 1995: 197–198) or the *Gen* operator (Chierchia 1995: 210). The idea that both stand in a local licensing relation to *Hab*_[+Q] is inspired by the analysis of NPIs (Ladusaw 1979), which effectively makes genericity a polarity phenomenon (Chierchia 1995: 212ff.). The main innovation is the analysis of ILPs as inherent generics, capitalizing on the assumption that polarity phenomena are rooted in the lexicon (with effects on sentences). The lexical entries of ILPs have an abstract habitual morpheme *Hab*_[+Q], whereby its agreement feature [+Q] can be locally licensed *only* by *Gen*, and never by overt Q-adverbs. ILPs are also taken to involve generic quantification over situations. Their Davidsonian situation variable *s* is bound by *Gen* making it unavailable for binding by overt Q-adverbs. Aside from the problems with Chierchia’s (1995) proposal, some of which are detailed in Fernald (2000), the Czech morpheme *-va-* cannot be the spell-out of the covert functional head *Hab*_[+Q]. If it were, it should have the same semantic properties as its covert counterpart in the logical representation of primary and secondary imperfectives on their generic interpretation. However, it does not. Moreover, if *-va-* were the spell out of *Hab*_[+Q], it should be uniformly incompatible with all ILPs, because their *Hab*_[+Q] can be locally licensed *only* by *Gen*, and never by overt Q-adverbs. However, *-va-* is compatible with ILPs provided they allow for an “interruption” construal, which also sanctions their co-occurrence with overt Q-adverbs (see Section 4.1.1).

the proper sense), be they formally unmarked or marked by means of dedicated markers like the Czech *-va-*, have modal import and predictive force, precisely because they express some habit or regularity. As such they have implications for other accessible worlds, than just the actual world, and perhaps all possible worlds. The modality of habitual sentences is not only compatible with their actualization entailment, or better realization inference, but is their *sine qua non* semantic property. Hence, a purely temporal analysis in terms of quantifiers that strictly quantify over times in the actual world will not do to capture the meaning of habitual sentences.

Finally, there is another problem faced by Cable's proposal. Cable (2022) assumes that any modality of habitals must be contributed by imperfective morphology, or some other material in the sentence, which has the meaning of a modal operator quantifying over alternative worlds other than the actual one, but never by perfective morphology, which is assumed to be non-modal. However, not only is there no notional and formal incompatibility between perfectivity and genericity as well as habituality (specifically for Slavic languages see Filip 1993b, 1994, Filip & Carlson 1997, and elsewhere), which Cable does not deny, but also perfective forms when used to express generic and habitual statements possess a modal semantics, and so necessitate quantification over alternative worlds other than the actual one. For instance, the English generic sentence (8a), which has a purely intensional interpretation, and is true despite having no actualized instances that count as evidence for its truth, and may never have any, is naturally rendered with a perfective verb in Czech:

- (16) Po viceprezidentovi *nastoupí*^{PFV} na místo prezidenta předseda
after vice.president ascend.PRS.3SG on place president chairman
sněmovny.
assembly
'The Speaker of the House succeeds the Vice President.'

This leads me to make some observations on the use of perfective forms in generic (and habitual) sentences, as a conclusion to this section. Some examples of generic sentences with naturally occurring perfective forms from Slavic languages, Russian, Serbo-Croatian and Czech, as well as Romance (Italian) are given below:²³

²³The Czech example (17c) is taken from <https://www.kosmas.cz/oko/ukazky/430121/o-panovnici-elite-prezidentu-povysenci-a-polarnikovi-v-nervoznim-pohybu/>, accessed July 9, 2023.

- (17) a. *Nastojaščij drug vseгда pomožet^{PFV}.*
 real friend always helps
 ‘A real friend will always help you.’ (a Russian proverb)
- b. *Svako jutro popijem^{PFV} čašu rakije.*
 every morning PREF.drink glass brandy
 ‘Every morning I drink a glass of brandy.’
 (Serbo-Croatian; Mønnesland 1984: 62)
- c. *To se jeden nalítá^{PFV} v úřadovnách prezidentského paláce.*
 it REFL one PREF.fly in offices president palace
 ‘What a runaround one must take in the offices of the presidential palace.’ (Czech)
- d. *Sempre, quando mi vide^{PFV}, il custode aprì^{PFV} la porta.*
 always when me saw the janitor opened the door
 ‘Always when he saw me, the janitor opened the door.’
 (Italian; Bonomi 1997: 508)

Dahl (1995) claims that the use of perfective forms in generic (and habitual) sentences in Slavic languages is another “case of deviant behavior” (Dahl 1995: 420) of their aspect systems. One must wonder about the justification for this claim, given that perfective forms are commonly used in generic sentences in Romance languages, for instance, as the above example from Italian shows. What is noteworthy though is that perfective forms are commonly used in generics and habituals in West Slavic languages.²⁴ In East and South Slavic languages they are less common (Forsyth 1970, Fortuin & Kamphuis 2015, Wiemer & Seržant 2017), but perhaps not as rarely used as is generally thought (Wiemer 2023).²⁵

Not only is perfectivity notionally compatible with genericity (and habituality), but also their respective overt markers are compatible, because they can co-occur on the same verb form: e.g., in Lithuanian “frequentative past” (Dambriunas et al. 1966), or in Awa (New Guinea) (Loving & McKaughan 1964).

²⁴For a recent study on the uses of the perfective present in Polish generic (and habitual) sentences see Wiemer et al. (2025).

²⁵In West Slavic language the use of perfective forms for the expression of generic and habitual statements is common, though their frequency as well as the functional range depends on the type of generalization to be expressed, and lexical idiosyncracies of the relevant (im)perfective forms. For instance, in Czech, perfective verbs are common or even preferred in cooking recipes, assembly instructions, game rules like rules of chess, proverbs, to name just a few types of generic statements.

In sum, genericity (and habituality) is orthogonal to the imperfective and perfective distinction (see also Filip & Carlson 1997). Acknowledging this separation of grammatical aspect from genericity (and habituality) is not to deny that there are close semantic affinities between imperfectivity and genericity, as well as habituality, as its special case.²⁶ Moreover, acknowledging that genericity (and habituality) is orthogonal to the imperfective and perfective distinction means that markers like the Slavic morpheme *-va-* that systematically enforce only a generic (or a habitual) interpretation of sentences need not be subsumed under imperfective aspect.

In support of this point, in the next section, I show that the Czech generic morpheme *-va-* contributes semantic and pragmatic properties to the meaning of generic sentences that their corresponding sentences headed by non-generic imperfective verbs, both primary and secondary, do not have. This also supports my conclusion that generic *-va-*marked verbs do not neatly fit the Slavic imperfective paradigm, and “general imperfective” paradigm in natural language (in the sense of e.g., Comrie 1976).

4 Semantic properties of the Slavic generic morpheme

4.1 When the generic morpheme *must*, *must not* and *may* be used

This section identifies cases in which verbs marked with the generic morpheme *may*, *must* and *must not* be used in contrast to imperfective verbs, which are not formally marked for genericity with this morpheme. The examples come from Czech. However, if a given Slavic language has a verb with the generic marker and could be used in the same context as that of a corresponding Czech verb, it is predicted that the same observations should hold for it that are made for Czech.

4.1.1 When the generic morpheme **MUST NOT** be used

The generic morpheme *-va-* must not be used in episodic contexts, where the reference is (i) to a particular on-going situation (progressive) or (ii) to a closed set of plural situations at a given world and time; neither can it be used (iii) for the expression of generalizations that permit no exceptions, nor (iv) for the

²⁶For instance, aspectually stative predicates in Slavic languages are expressed by imperfective verbs, and aspectual stativity is a key semantic property of generic predicates. Moreover, from the grammaticalization point of view, the expression of genericity (habituality) is tied to markers that become grammaticalized on the path from (present) progressive markers to imperfective ones (Bybee & Dahl 1989).

expression of generalizations with unrealized particulars in the actual world, i.e., in only dispositional, purely hypothetical generalizations.

4.1.1.1 Lack of reference to a particular (on-going) situation

One of the well-known tests tapping into the semantics of the Slavic perfective/imperfective distinction is the possibility of reference to a particular episodic situation that is on-going at a specific moment of time, i.e., the possibility of the progressive interpretation. All imperfective verbs that denote stage-level predicates pass this test, but those denoting individual-level (stative) predicates do not, and neither do perfective verbs.

Most importantly, all verbs marked with the generic morpheme *-va-* also fail this test. In traditional Bohemistics, this property is well-known as their hallmark property, which is referred to as the “non-actuality” or “atemporality” property at least since Kopečný (1948). (See also Kopečný 1962, 1965, Kopečný 1966: 259.) The example below, which is adapted from Kopečný (1948), shows that Czech verbs marked with *-va-* cannot be felicitously used in answers to questions about what is *actually* going on ‘now’ at the speech time. In contrast, their imperfective counterparts without the morpheme *-va-* can. It is important to emphasize that the evaluation with respect to “extended-now” that covers extended intervals of time is irrelevant for this test:

(18) *Context*: ‘What are you doing right now?’

- a. # Psávám jeden naléhavý dopis.
 write.GEN.1SG.PRS one well-known letter
 ‘I write one urgent letter.’ (as a rule/as a habit/often/rarely/usually)
- b. Píšu^{IPFV} jeden naléhavý dopis.
 write.1SG.PRS one urgent letter
 ‘I am writing one urgent letter.’

The generic morpheme *-va-* if it is attested on a verb in other Slavic languages will also give rise to the oddity that we see in (18a).

The semantic property of “non-actuality” or “atemporality” contributed by the Czech morpheme *-va-* along with its productivity and compositional transparency (see also Section 3) lead Kopečný (1948) to propose that it is a grammatical marker of a “third aspect”, namely an “atemporal aspect” that is independent of the Slavic perfective/imperfective distinction (see also Kosek 2014). As a wide-reaching consequence, Kopečný (1948) rejects the claim made by Koschmieder

(1929) that Indo-European languages lack a grammatical category of “atemporal-ity”.²⁷ On Kopečný’s third “atemporal aspect” hypothesis, the Czech verb system might be schematically represented as in Figure 2, albeit recast in terms of the non-generic/generic distinction, as it is used in the contemporary literature and in this chapter.

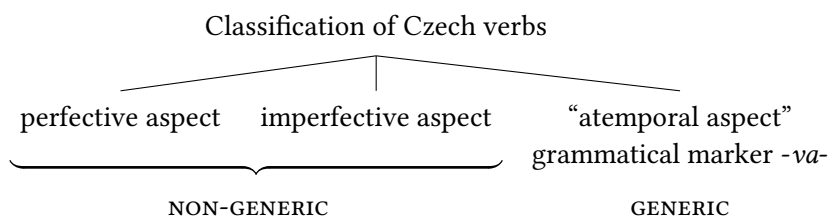


Figure 2: Classification of Czech verbs (following Kopečný 1948 and elsewhere)

Aligned with Kopečný’s “third atemporal aspect” hypothesis, Lamprecht et al. (1986: 325) states that Czech verb forms that are built with the generic morpheme *-va-* (‘generic’ in my terms) are clearly not imperfective forms, because imperfectives are formed by other morphologically means. Also Kosek (2014: 152) observes that in Czech the “third aspect” with its dedicated formal expression by means of the morpheme *-va-* has been established since the 16th century, when it became independent of the perfective/imperfective distinction. However, Kosek, unlike Kopečný, erroneously views it as a means for the expression of iterativity, or frequentativity.

Kopečný’s (1948) “third atemporal aspect” hypothesis is of interest here in so far as it highlights the hallmark “atemporal” or “non-actuality” property of the Czech morpheme *-va-*, i.e., that it formally marks sentences that lack reference to a specific situation in all its occurrences. As observed above (see Section 2), it is one of the key properties of generic predicates in natural languages generally, which makes them aspectually stative. Kopečný’s account of the Czech morpheme *-va-* is a notable improvement on its traditional treatment as an imperfective marker carrying the iterative or multiplicative feature, which it quite clearly lacks (as will be shown below).

²⁷Kopečný (1948): “Není tedy zcela správné tvrzení Koschmiedrovo v jeho Nauce o aspektech (33), že indoevropské jazyky nemají [153] gramatickou kategorii atemporalnosti (mimočasovosti, “pozaczasowości”).” [Koschmieder’s claim that Indo-European languages have no grammatical category of atemporality (extra-temporality, ‘pozaczasowość’ [Polish]) is not entirely correct. (my own translation)]

In sum, based on the above observations, the first key property of Czech verbs that are formed with the generic morpheme can be stated as follows:

- (19) SEMANTIC PROPERTY 1 (to be revised): The generic morpheme *-va-* derives verbs that systematically lack reference to particular situations. It is a property at the core of stativity, and one of the properties of generic predicates in natural languages.

Semantic Property 1 in (19) clearly separates the generic morpheme from the imperfectivizing suffix in Slavic languages, which formally marks secondary imperfectives that, as a whole class, are not aspectually stative. Just like primary imperfectives, such as *píšu* ‘I am writing’ in (18a), secondary imperfectives freely alternate between episodic and generic readings, and can be used with reference to a specific situation that is on-going at some given reference time, as the following attested example shows:²⁸

- (20) V křoví právě přepočítával^{IPFV} bankovky v peněžence,
 in shrubs just PREF.COUNT.IPFV.PST.3.SG banknotes in wallet
 strážníky na sebe upozornil světlem z mobilu.
 policemanPL.ACC on himself make.awarePST.3.SG light from mobile
 ‘At that moment he was in the shrubs counting money in his wallet, and
 the light of his mobile phone alerted the police to him.’

4.1.1.2 Incompatibility with iterativity, or a mere plurality of situations

In general, the lack of reference to a specific situation, which is a hallmark property of generic predicates (related to their stativity), is closely tied to their incompatibility with iterative adverbials (or cardinal count adverbials) like ‘three times’ or ‘several times’. Consider the following example:

- (21) # Včera Pavel hrával třikrát šachy s dědou.
 yesterday Paul play.GEN.PST three.times chess with grandpa
 #‘Paul used to play chess with grandpa three times yesterday.’

The oddity of the above sentence must be due to the clash between the generic marker *-va-* and the iterative adverbial ‘three times’ which counts three episodes that occurred within a single interval of time ‘yesterday’. That is, ‘yesterday’

²⁸The example is from [https://www.praha11.cz/redakce/tisk.php?lanG=cs&clanek=9892&slozka=7155&as4uOriginalDomain=www.praha11.cz&as4u_protocol=https&";](https://www.praha11.cz/redakce/tisk.php?lanG=cs&clanek=9892&slozka=7155&as4uOriginalDomain=www.praha11.cz&as4u_protocol=https&) accessed on August 15, 2025.

enforces an episodic reading of a sentence, which must be evaluated with respect to a particular world/time, but the generic morpheme *-va-* requires evaluation at alternative possible worlds than just one (actual) world.

The generic morpheme *-va-* can scope over the iterative adverbial, if the context does not indicate otherwise, as we see (22). Here the closed sequence of iterated three eventualities forms the basis of the generalization which is a part of a larger pattern that holds every day, and across different possible worlds.

- (22) Psával mi aspoň třikrát každý den.
write.GEN.PST me at.least three.times every day
'He used to write to me at least three times every day.'

In (21), however, 'yesterday' restricts the reference to a single interval/world which excludes a generic interpretation of (21), and consequently the use of the generic morpheme is odd. The corresponding non-generic sentence (23) without *-va-* is perfectly acceptable with the iterative adverbial 'three times'. The main verb here is a non-generic primary imperfective verb:

- (23) Včera Pavel hrál^{IPFV} třikrát šachy s dědou.
yesterday Paul play.PST three.times chess with grandpa
'Paul played chess with grandpa three times yesterday.'

Secondary imperfective verbs formed with the imperfectivizing suffix (IPFV), which are also non-generic like primary imperfectives, are compatible with iterative adverbials like 'three times':

- (24) Sice jsem již třikrát vyhrávala^{IPFV}, ale nikdy jsem
while AUX already three.times PREF.play.IPFV.PST but never AUX
nevyhrála^{PFV}.
NEG.PREF.play.PST
'Although I was already winning / was about to win three times, I never won.'

The incompatibility with iterative adverbials is a test that generally separates verbs with the generic morpheme *-va-*, on the one hand, from primary and secondary imperfective verbs, on the other hand; only the latter can co-occur with iterative adverbials, modulo their lexical semantics and context in sentences that describe episodic situations that are not a part of a larger pattern. This also means that verbs with the generic morpheme *-va-*, on the one hand, and secondary imperfective verbs with the imperfectivizing suffix (IPFV) along with primary imperfectives, on the other hand, make different contributions to the meaning of

a sentence. This provides one piece of evidence in support of separating verbs with the generic morpheme *-va-* from primary and secondary imperfectives.

Generally, the incompatibility of a morpheme with iterative adverbials means that it cannot be assimilated to markers in natural languages that are variously labeled as iterative, multiplicative, frequentative, repetitive, pluractional, and the like, of the kind that are used to encode a mere plurality of situations (i.e., a closed series of particular episodes or situations of the same type at a given world and time). The incompatibility of the generic morpheme *-va-* with iterative adverbials clearly shows that its common labels “iterative” or “multiplicative” and the like are misnomers at best (see also Bílý 1986), or a category error at worst (*pace* standard Czech reference grammars, e.g., Petr 1986, Kosek 2014, Lamprecht et al. 1986, Nübler 2017 or Dahl 1995).

Based on the above data and observations, we may update Semantic Property 1 in (19) as follows:

- (25) SEMANTIC PROPERTY 1 (final version): The generic morpheme *-va-* derives verbs that lack reference to particular situations. It is a property at the core of stativity, and one of the properties of generic predicates in natural languages.

Corrolary: The generic morpheme *-va-* derives verbs that are incompatible with iterative adverbials like ‘three times’, ‘several times’ used to denote a cardinality of a closed set of situations at a given time interval and world (i.e., a set that is not a part of a larger pattern, or generalization).

Semantic Property 1 is related to independent arguments made elsewhere that genericity (and habituality) is clearly separate from iterativity, and the two must not be conflated and confounded, both on formal and notional grounds (see e.g., Carlson 2012a).²⁹ Iterativity involves a closed set of a plurality of events, eventualities, situations, or cases of the same type (or a plurality of time intervals associated with them), but this has virtually nothing to do with habituality and genericity, i.e., with what gives rise to a generalization, a regularity, or a pattern (see e.g. Krifka et al. 1995, Filip 1993b, Carlson 2005, 2008, 2012a, i.a). It is neither sufficient nor necessary for the truth of habitual and generic statements. For instance, a mere plurality of situations is not sufficient, because it can be a part of the meaning of what clearly are non-generic statements, be it a sequence of temporally close situations that constitute a single complex situation (e.g., *Last night, he knocked three times on the door*), or multiple separate situations (e.g.,

²⁹See also e.g. Bertinetto & Lenci (2012) for an overview of notions related to iterativity and their delimitation from genericity and habituality.

During our train ride to Rochester, the conductor came to our car three times to check our tickets), which may also be spread over a large interval (e.g., *During his lifetime Franz met Milena only twice*). A closed set of a plurality of situations of the same type is not necessary for the truth of generic statements, because some have a purely intensional interpretation with no actualized verifying particulars, as in (8a). Moreover, there are generic statements with ILPs whose truth is not inferable from any array of a particular situations at all, such as *Bob owns a yacht* that presupposes one situation of acquisition like buying (Krifka et al. 1995: 38), or *Bob is a bachelor* for which it makes no sense to try to identify particular situations that count as evidence for its truth (Carlson 1995). Such observations also indicate that genericity and habituality should not be confounded in analyses of phenomena referred to as iterative aspect, event-internal or event-external iterativity, frequentativity, pluractionality, multiplicativity, iterated semelfactivity and the like (*pace* e.g., Nef 1986, Van Geenhoven 2001, 2004, Scheiner 2003).

The conflation of iterativity (i.e., a mere plurality of situations) with habituality and/or genericity is also common in the analyses of habitual and generic readings of imperfective sentences. A common strategy is to add to the imperfective operator a parameter capturing the number requirement on its application. Examples of this strategy can be found in Ferreira (2016) (also Ferreira 2004, 2005, and elsewhere), and Hacquard (2006), for instance. Imperfective forms *either* denote singular eventualities under their episodic progressive interpretation, *or* a plurality thereof under their habitual/generic interpretation.³⁰ For the latter, Ferreira (2005: 29) assumes a phonologically null “habitual/generic operator *hab*” when they occur without an overt Q-adverb or restrictor. Both the episodic progressive and habitual/generic interpretations share the same temporal component, namely existential quantification over time intervals, and also the same modal component with modal quantification over possible worlds, adapted from the analyses of the progressive aspect (as developed in Dowty 1979, Landman 1992, Portner 1998). In short, the habitual/generic operator *hab* is the plural counterpart of the progressive operator *Prog*, where *Prog* is defined on singular (atomic) eventualities. The *Prog* truth conditions are stated with respect to a pair of an interval and a world $\langle i, w \rangle$, iff there is a singular eventuality *e* in *w* and modal continuations in possible worlds in which it terminates or is completed. A habitual (generic) sentence like *John plays soccer (regularly)* is true in all the worlds in which there were plural eventualities of John playing soccer before the utterance time, and there is an ongoing plural eventuality of John playing soccer at some

³⁰ Assuming that VPs denote sets of time intervals, and VPs like NPs can be singular or plural, as e.g. Ferreira (2016: 358) proposes, the progressive (episodic) interpretation of a VP amounts to singular time intervals at which it holds and the “habitual” (generic) interpretation to plural time intervals.

reference time, whereby all potential external obstacles to the continuation of this plural eventuality are removed.

Setting aside that a proper characterization of the modal continuation component of Prog is still outstanding, and that the meaning of habitual/generic sentences cannot be analyzed in terms of a plurality of events (or plurality of time intervals), one obvious problem with this proposal is that it does not cover sentences with a purely dispositional reading like *Mary handles the mail from Antarctica*, which may be true even if Mary never handled any mail from Antarctica before the utterance time, and such mail may never ever arrive in the future. Neither can it handle all the relevant aspects of the meaning of bare habitual sentences like *John plays soccer*. This sentence may mean that John is not averse to playing soccer, he has an ability or disposition for playing soccer, and I can truthfully assert it, based on my experience of having seen him play soccer once in the past. This allows me to conclude that if John were in a situation to play soccer, he just might likely play soccer. *John plays soccer* is true, even if there are no multiple occasions of John playing soccer at some reference interval including the reference time, and John may never play soccer in the future. Moreover, suppose you asserted that John does not engage in any sports activity. I can contradict you by uttering *John plays soccer*, even if I have only seen John play soccer once in the past. But this cannot be captured if we assume that “habituals are about plural events” (Ferreira 2016: 353) combined with the modality that has been associated with or is akin to Prog.

4.1.1.3 Generalizations permitting no exceptions

One of the key properties of generic sentences is that most are exception-tolerating, but are also compatible with no exceptions (see Section 2). The generic morpheme, as I argue, has a meaning that combines a generalization assertion with an exception or a counterexample inference that there are or may be exceptions to the generalization. *Exceptions* to the generalization are compatible with its truth, while *counterexamples* are incompatible with its truth (see e.g. Nickel 2010). This property was already discussed by Filip (1993b, 1994, 2019, 2021), and here, it is captured as follows:

- (26) SEMANTIC PROPERTY 2: The generic morpheme *-va-* contributes the uncancelable inference (as part of its truth-conditional content) that there are actual or possible exceptions or counterexamples to the generalization.

Corollary: The generic morpheme *-va-* is incompatible with universal generalizations (e.g., universal laws of nature), or any generalization that is

commonly known to permit NO EXCEPTIONS, possibly in ALL THE KNOWN CONDITIONS.

The clearest examples of Semantic Property 2 (in (26)) are sentences in which the generic morpheme *-va-* clashes with universal laws of nature (as in (27a)) or definitional statements that necessarily hold for all the members of a given kind (as in (27b)). Such exceptionless regularities call for verb forms without the generic morpheme, which, in the Czech examples below are primary imperfective verbs.

- (27) a. Země se {točí / #točívá} kolem slunce.
 earth REFL revolve.3SG.PRS revolve.GEN.3SG.PRS around sun
 (Intended:) 'The Earth revolves around the Sun.'
- b. Valčík {je / #bývá} ve třetíčtvrtečním taktu.
 waltz be.3SG.PRS be.GEN.3SG.PRS in three.quarter time
 (Intended:) 'A waltz is in three quarter time.'

Semantic Property 2 is also clearly manifested in the incompatibility of the generic morpheme *-va-* with an overt universal quantifier in the same clause:

- (28) a. {#Každou sobotu / #vždycky} sedává Honza v hospodě.
 each Saturday always sit.GEN.3SG.PRS Honza in pub
 'Every Saturday / #always Honza usually tends to sit in a pub.'
- b. {Každou sobotu / vždycky} sedí Honza v hospodě.
 each Saturday always sit.3SG.PRS Honza in pub
 'Every Saturday / always Honza sits in a pub.'

(28a) is at least odd, if not involving a contradiction (but see below analogous examples (29a) and (29b), where there is no oddity), assuming that the universal Q-adverbs, *každou sobotu* 'every Saturday' and *vždycky* 'always', are used with their universal quantificational force. If so, they require that the sentence be true just in case Honza's sitting in a pub occurs on each and every Saturday with no exception, or on each appropriate contextually determined occasion. However, the generic morpheme *-va-* contributes the uncancellable inference that this precisely is *not* necessarily the case. That the oddity of (28a) must be due to the generic morpheme *-va-* is evidenced by the fact that (28b) is perfectly acceptable and true, because it minimally differs from (28a) by lacking *-va-*.

However, the generic morpheme *-va-* may co-occur with universal Q-adverbs, if they are used for "intensification of the strength" of the expressed regularity (Danaher 2003: 45), rather than with their inherent universal quantificational force and so allowing no exceptions. The examples below are taken from Danaher (2003: 45):

- (29) a. Mládež ve Vídni se zabývala Hebblem – já jsem vždycky
 youth in Vienna REFL concerned Hebbel I AUX always
býval skeptický k takovým módním proudům.
 was.GEN skeptical to such fashion streams
 ‘Viennese youth were all reading Hebbel – I was usually always
 skeptical about these fashionable influences.’
 (from K. Čapek: *Hovory s TGM*)
- b. „Je to divný,“ pokračovala pak rychlým a věcným šepotem, „
 is it strange continued then quick and matter.of.fact whisper
 jeden šuplík má zamčenej, a nikdy ho nemívá zamčenej.
 one drawer has locked and never.NCI it NEG.has.GEN locked
 A nepasuje mi do něj žádný klíč.“
 and NEG.fit me into it no.NCI key
 “It’s strange,” she continued in a quick and matter-of-fact whisper,
 “one of his desk drawers is locked and he never has it locked. And
 none of my keys fit the lock.” (from H. Bělohradská: *Poslední večere*)

The felicitous co-occurrence of the generic morpheme *-va-* with universal Q-adverbs is similar to the felicitous use of non-universal Q-adverbs together with universal Q-adverbs in combinations like *usually always* or *usually never* in English, where the universal Q-adverbs also arguably lack their inherent universal quantificational force, and function as intensifiers of a regularity that is known to have exceptions:

- (30) a. I am usually always happy, but today I feel really depressed.
 b. I am usually never neurotic about being messy and keeping things tidy, but I can’t seem to go to sleep if clothes are hanging up to dry in my room.
 (Barbara Partee, p.c.)

Semantic Property 2 in (26) also predicts that the generic morpheme *-va-* will be unacceptable, or at least strongly dispreferred, for the encoding of properties that are denoted by inherent individual-level predicates like *know (French)*, *believe*, *have long arms*, *be intelligent* that are stable for an individual, in so far as they spread over a relatively large interval of that individual’s life-time, and possibly over most/all of its life-time, and including all of its subintervals and moments. One way of thinking about this property is that individual-level predicates involve a kind of implicit universal quantification over intervals and moments at which they are true of individuals. This motivates why the generic morpheme *-va-* cannot (freely and straightforwardly) apply to imperfectives that denote

individual-level predicates: precisely because it introduces the uncancellable inference that there are or may be exceptions or counterexamples, i.e., that the characterizing property does not stably hold of an individual at all the moments of its life-time. This is illustrated by the following Czech example:

- (31) Je rodilá Francouzka, a proto #znává francouzsky.
 is born French.woman and therefore know.GEN.3SG French
 (Intended:) ‘She is a French native, and so knows French.’

Similarly, *know (French)* in English is odd with a Q-adverb like *sometimes*, as we see in (32a). In contrast, *sometimes* is compatible with the individual-level predicate *be a California resident*, as it can be construed as describing a property holding with interruptions, on and off at different subintervals of evaluation, sometimes it holds, and sometimes it does not. This fluctuation in truth values generates “gaps” or “interruptions” in their application and hence also a plurality of separate cases, which is a prerequisite for the application of the Q-adverb (see e.g. Fernald 2000 and related suggestions by de Swart 1991, i.a.):

- (32) a. # Mary sometimes knows French.
 b. Max is sometimes a California resident. (Fernald 2000)

The possibility of an “interruption” construal also makes the combination of the generic morpheme *-va-* with individual-level predicates acceptable, and so Semantic Property 2 in (26) is satisfied. Some Czech examples are given below, where (34b) is taken from *Seven poems written on a mirror* by Karel Kryl (1976):

- (33) a. myslit (si) ‘to think’ → myslívat (si) ‘(to tend) to think’ (see (34a))
 b. patřit ‘to belong (to)’ → patřívát ‘(to tend) to belong to’ (see (34b))
 c. věřit ‘to believe’ → věřívát ‘(to tend) to believe’
 d. mít ‘to have’ → mívat ‘(to tend) to have’
- (34) a. Každopádně nejste sám, kdo si to někdy myslívá.
 in.any.case NEG.be.2PL alone who REFL it sometimes think.GEN.3SG
 ‘In any case, you are not the only one who occasionally thinks like this.’
 b. ... občas i k slovesu milovat patřívá zájmeno zvrtné.
 at.times even to verb love belong.GEN.3SG pronoun reflexive
 ‘... the reflexive pronoun occasionally belongs even to the verb *love*’

The “interruption” construal of individual-level predicates, which facilitates their use in the scope of non-universal Q-adverbs in generic sentences, might look like a kind of coercion of individual-level predicates into a stage-level predicate interpretation. However, if this were a matter of coerced stage-level predicate interpretations, then they should be also available for individual-level predicates in episodic contexts, for instance, triggered by adverbials denoting particular spatio-temporal locations. However, such coerced stage-level predicate interpretations of individual-level predicates are not available:

- (35) #Max is a California resident right now in his garden.

Such considerations lead Carlson (1977), Kratzer (1989), and Fernald (2000) to propose that individual-level predicates in generic sentences like (32b) are not to be treated as having first undergone a shift into a stage-level predicate interpretation, but rather still have arguments that are of the individual sort, and denote individual-level stative properties, albeit having truth values that vary across evaluation times, are true on and off. *Mutatis mutandis* the same holds for individual-level predicates in the scope of the generic morpheme *-va-*.

Another manifestation of Semantic Property 2 in (26) is the oddity, or even unacceptability, of the use of the generic morpheme *-va-* in generic contexts in which what is at stake are characterizing generic properties, such as inherent dispositions, abilities, and acquired professions, that are understood as holding without interruptions for a given individual over an extended period of that individual’s life-time, and all its subintervals and moments. Consider the following contrast:

- (36) *Question*: ‘What is Mary’s profession?’
- a. Učí na střední škole.
 teach.3SG.PRS on middle school
 ‘She teaches at high school.’
 Inference: ‘Mary is a high-school teacher.’
 - b. # Učívá na střední škole.
 teach.GEN.3SG.PRS on middle school
 Intended: ‘She teaches at high school on and off / sometimes / often...’
 Inference: ‘Being a high-school teacher is not Mary’s profession.’

The question about Mary’s profession is best answered with (36a), which is headed by the imperfective non-generic verb *učí* ‘she teaches’, here under its

generic interpretation, and is paraphrasable with ‘Mary is a high-school teacher.’ Teaching as one’s profession typically presupposes having acquired the requisite training and accreditation, which then holds for an extended interval of one’s life, and all its subintervals, without interruptions, including times off work (week-ends, holidays, or absence due to illness).

In contrast, (36b) which is formally marked with the generic *-va-*, would be odd or infelicitous in this context, because it conveys the inference that teaching only holds with interruptions of Mary. This invites the inference that Mary only teaches on and off, because she may be a temporary or a substitute teacher, and may also have other alternative jobs or occupations. In short, it invites the inference that Mary is not a teacher by profession.

4.1.1.4 Generalizations with unrealized particulars in the actual world

The generic morpheme *-va-* cannot be used in generic sentences that concern unrealized rules, dispositions, functions, instructions and the like.³¹ Consider the following example from Czech:

- (37) *Scenario*: We have just bought a new machine for crushing oranges, but we have not yet used it, and it is still packed in its original box.

Tento stroj drtí / #drtívá pomeranče.
this machine crush.3SG.PRS crush.GEN.3SG.PRS oranges
(Intended:) ‘This machine crushes oranges.’

What is at stake under this scenario is the intended function of the specific machine token in the domain of discourse, which is as yet not realized. Here, only a verb that sanctions a purely intensional (dispositional) interpretation can be used. In Slavic languages, it is a non-generic verb, which, in the example above, is the primary imperfective verb *drtí* ‘it crushes’. The generic verb marked with the generic morpheme makes the whole sentence false under the given scenario. The reason for it is that it requires that this particular machine have already been put to use, and so there have been (observed) instances of it crushing of oranges in the actual world from which the expressed generalization is inductively

³¹In the studies on genericity and habituality this meaning component is commonly (though not uncontroversially) referred to as the “actuality entailment”, which was originally coined by Bhatt (1999) to describe the implicative inference that arises with ability modals in perfective aspect. However, in the context of habitual/generic sentences it is perhaps not appropriate, because the requirement that there be actual instantiations of the generalization tends to survive in negated habitual/generic sentences. Therefore, henceforth, I will use the term *realization inference*.

projected. We may formulate this realization constraint on the use of the Slavic generic morpheme as follows:

- (38) SEMANTIC PROPERTY 3: The generic morpheme *-va-* carries the uncancellable inference (as part of its truth-conditional content) that there are verifying instances in the actual world that count as evidence for the truth of generic sentences it marks.

The realization inference that is often taken to be a signature property of so-called “habitual” sentences is also mistakenly correlated with their lack of modality (intensionality), and used as an argument for not treating “habitual” sentences as a special case of generic sentences which are inherently modal, as all agree. However, as many semanticists also agree, any sentence that describes a habit, regularity, pattern, generalization of any kind has a modal (intensional) import (see also Section 2 above, and references therein). This is because its meaning transcends what we observe in the actual world; they are not only about what was realized, but also predict what is possible or likely. By virtue of having this predictive, law-like force, habitual sentences also have implications for other accessible alternate worlds, rather than just the actual world, and perhaps all possible worlds. Consequently, so-called “habitual” markers also introduce quantification over alternate possible worlds.³² If the alleged lack of this modal component is the reason for not treating so-called “habitual” markers as falling under genericity, or as not qualifying as generic markers, then this exclusion is in error.

We can illustrate this point with the following example from Czech:

- (39) Děti v této škole mívají dobré známky.
 children in this school have.GEN.3PL.PRS good grades
 ‘Children in this school tend to have good grades.’

The truth of (39), which contains the generic marker *-va-*, is not just based on a mere plurality of situations of various children having already actually received good grades, but (39) also gives rise to counterfactual inferences that transcend observable real world conditions we have access to: namely, for instance, if a child were to attend this school, that child too would most likely or probably get good grades. The reason for this is that the truth of (39) also depends on there being some underlying reason(s) or cause(s) for this, such as the quality of instruction, parents’ attention, their innate predispositions, etc. While we may not know the

³²Similar observations are also made by Boneh & Doron (2010) in connection with the Modern Hebrew periphrastic “habitual” construction with *haya* (‘was’)+participle.

particular reason(s) or cause(s), whatever they might be they also ground the truth of (39), besides our observations of children in this school getting good grades.

The same considerations that motivate the intensionality of (39), which has an overt marker of genericity, also motivate the intensionality of *Children in this school have good grades* in English, which lack any overt generic marker (see e.g., Pelletier & Asher 1997 for similar examples), and which are associated with a logical representation with the intensional generic operator GEN (in Pelletier & Asher 1997 or Krifka et al. 1995, i.a.). If one accepts that the English sentence *Children in this school have good grades* has an intensional (modal) import, as most do, then one should acknowledge that (39) does, as well.

That the realization inference of generic and habitual sentences, including those that have overt so-called “habitual” markers (such as we find in Tlingit or Modern Hebrew), is compatible with the intensionality (modality) which is characteristic of sentences expressing regularities, should not be surprising given that there are other modal operators that introduce the realization inference (see Bhatt 1999 and Hacquard 2006).

4.1.2 When *-va-* MUST be used: Positive counterinstances

Due to its exceptive component, here captured by Semantic Property 2 in (26), the addition of the generic marker *-va-* to a generic sentence may reverse its truth conditions. This also supports its treatment as part of the lexical meaning of *-va-*. A case in point is its use in generics that have counterexamples that count as positive counterinstances (in the sense of Leslie 2008) to the generalization. Generalizations that allow for exceptions partition the domain of their applicability into confirming cases and non-confirming cases. However, not all non-confirming cases are alike, not all render such generalizations false. As Leslie (2008) shows, for a generalization to be true (or false) it matters how exactly non-confirming cases to it are judged to fail it. Consider the following contrast:

- | | | |
|------|-------------------------|-------|
| (40) | a. Birds fly. | TRUE |
| | b. Books are paperback. | FALSE |

(40a) is an example of an exception-tolerating, non-universal generalization that is true by virtue of the majority of birds having the BIRD-kind characterizing property of flying, and it is not falsified by exceptional non-flying bird subkinds like penguins, ostriches or emus, or individual birds that do not fly because of some accidental defect. In contrast, (40b) is false, even if the majority of books are paperbacks.

As Leslie (2008) argues, we are likely to judge a generic sentence to be true, if exceptions to it constitute “negative counterinstances”. Non-flying birds constitute negative counterinstances to the positive property of flying characterizing the kind BIRD. They simply lack this positive property without providing any alternative positive property that would be equally (subjectively) salient or striking as flying and that we could pit against it. This makes such negative counterinstances easy for us to ignore, or set aside, when judging a generalization like (40a) as true.

In contrast, if exceptions to a generalization are viewed as positive counterinstances, we are likely to judge that generalization as false. (40b) is judged false, because exceptions to it, even if they are fewer than the confirming paperback members of the kind BOOK, do not simply fail to be paperbacks, but possess a positive distinctive feature, such as being hardback books. I.e., they have an alternative positive property that is subjectively for us just as salient, memorable, significant, or striking as having the positive property of paperback. Positive counterinstances are cognitively salient in our kind-related expectations, which makes them particularly hard for us to ignore when judging the truth of generics like (40b). Other examples of generics that are judged false due to positive counterinstances to them are: *Nurses are female* (there are also male nurses), *People have brown eyes* (there are people with blue, green, grey eye-color, for instance).

In order for a generalization with positive counterinstances to be true, it must contain some overt expression qualifying the generalization that signals that not all the members of the kind share the property that can be truthfully predicated of that kind. For instance, *typically* and *usually* do this qualifying job, as Leslie (2008) observes:

- (41) Typically/Usually, books are paperback. TRUE

Other overt qualifiers of generic statements include generic non-universal Q-adverbs like *usually*, *rarely*, *seldom*, quantity expressions like *in the majority of cases*, *in some cases*, spatio-temporal modifiers like *(books) on sale in this bookstore*, and also exceptives like *but*, prepositions *except (for)*, and ceteris paribus clauses.

In Czech, Slovak, and also to a lesser extent in Polish, in addition to the above restrictors, we can also use the generic morpheme *-va-* to qualify a given generalization and so guarantee its truth, which otherwise would be false. Taking examples from Czech, this can be illustrated by the contrast between (42a), on the one hand, and (42b) and (42c), on other hand:

- (42) a. Knihy jsou^{IPFV} brožované. FALSE
 books be.3PL.PRS paperback
 ‘Books are paperback.’

- | | | |
|----|---|------|
| b. | Knihy jsou ^{IPFV} {obyčejně / nejčastěji} brožované.
books be.3PL.PRS usually most.often paperback
'Books are {usually / most often} paperback.' | TRUE |
| c. | Knihy bývají brožované.
books be.GEN.3PL.PRS paperback
'Books tend to be paperback.' | TRUE |
| d. | Knihy bývají {obyčejně / nejčastěji} brožované.
books be.GEN.3PL.PRS usually most.often paperback
'Books are {usually / most often} paperback.' | TRUE |

(42a) contains the primary imperfective verb *jsou* meaning 'they are', and it is judged false based on the same reasoning about exceptions qua positive counterinstances proposed by Leslie (2008) for the English generic sentence (40b). Adding the Q-adverbs 'usually' or 'most often' in (42b) or the generic morpheme *-va-* in (42c), reverses the truth value to true. Q-adverbs and the generic morpheme can co-occur in the same clause, as (42d) shows (and I will return to this point further below).

The contrast between (42a) and (42c) shows that Slavic imperfective verbs have no exceptive meaning component that could function as a safeguard for the truth of the generalization, and the presence of *-va-* is sufficient to make a generic sentence true. This is yet another difference between verbs with the generic *-va-* and imperfective verbs. In (42a), we have a primary imperfective verb. Just as a primary imperfective so a secondary imperfective with the imperfectivizing suffix is insufficient to guarantee the truth of "bare" generics with positive counterexamples. Generally, an exceptive meaning component is not a part of the semantics of imperfective forms in natural languages. In this respect verbs with the generic morpheme *-va-* do not neatly fit the imperfective paradigm.

4.1.3 When *-va-* MAY be used: Pragmatic inferences based on quantity

The exceptive component of the generic marker *-va-*, as given in (26), also has pragmatic consequences in contexts when it *may* be used. Consider the following pair of sentences which only differ in their main verbs, the generic one marked with *-va-* and the related morphologically simpler non-generic verb:

- (43) a. Výuka začíná^{IPFV} od 8.00 a končí v poledne.
teaching start.3SG.PRS from 8:00 and ends in noon
'Classes start at 8 a.m. and end at noon.'

- b. Výuka začínává od 8.00 a končí v poledne.
 teaching starts.GEN.3SG.PRS from 8:00 and ends in noon
 'Classes tend to start at 8 a.m. and end at noon.' [as a rule, usually]

Suppose I registered for a course at school where classes are scheduled every day, and I ask the school administrator when classes start. Under this scenario, either (43a) with the non-generic verb *začíná* here meaning 'it starts' or (43b) with the generic verb *začínává* meaning 'it tends to start', 'it usually / often / as a rule / ... starts' can be used as an answer. Suppose the administrator responds with (43a). The next day I arrive at school shortly before 8 a.m., but my classes on this day in fact start only at 10 a.m., and so I waste two hours waiting. And yet, I could not blame the administrator for saying something blatantly false, because generic sentences like (43a) with a verb formally unmarked for genericity, on their generic interpretation, describe regularities that allow for exceptions, i.e., are exception-tolerating like most generic/habitual sentences (see Section 2). Indeed, (43a) can be continued without contradiction with ... *ale ne vždycky* '... but not always' or ... *s výjimkou pátku* '... except for Friday'.

While I could not blame the administrator for saying something blatantly false by using (43a), I could blame them for being misleading at least. Here is why. (43a) is also compatible with no exceptions, as many generic sentences are (see Section 2). Given that (43a) can be used to express an exceptionless generalization, i.e., the strongest generalization, from the point of view of the generalization strength, (43a) is the stronger alternative than (43b), which requires that there be actual or possible exceptions (Semantic Property 2). From the administrator's choice of (43a), the stronger alternative, when the weaker alternative (43b) with the generic verb could have also been used, I may infer the denial/rejection of the weaker alternative that requires that there be actual or possible exceptions. That is, (43a) can be understood as involving pragmatic strengthening to an exceptionless regularity, namely that classes always start at 8 a.m., without exception.

In contrast, in the described scenario, if the administrator utters (43b), I cannot not blame them for being misleading, or even giving me wrong information about when classes start. The reason for this is that the generic marker, besides its generic import, requires that there be actual or possible exceptions to the generalization, which excludes pragmatic strengthening of (43b) to an exceptionless generalization. Moreover, by using (43b) instead of (43a), the administrator implicates that they do not have enough information to be in the position to commit to the stronger alternative (43a), which allows a pragmatic strengthening to an exceptionless generalization.

Generally, the choice between the two lexical alternatives, a generic verb with *-va-* and a corresponding morphologically related non-generic verb

without *-va-*, gives rise to one of the two (defeasible) pragmatic quantity-based inferences which concern either the speaker's *certain*ty or *ignorance* about the existence or possibility of exceptions to it. In our example, the administrator may use the alternative with the generic morpheme, because they are *certain* that classes do not always start at 8 a.m., or that they are *uncertain* whether exceptions to their starting at 8 a.m. can be completely excluded. In the latter case, *-va-* may be used as a hedge to safeguard a generalization against refutation by states of affairs of which the speaker is ignorant, and so to ensure its plausible deniability.

This mechanism of quantity-based inferences is also exploited in legal discourse, and in the formulation of various rules, instructions, job descriptions, or customs, which includes generics that have a normative character. Below are some attested examples taken from Czech, a game rule (44a) and an institutional regulation (44b):³³

- (44) a. Zápas *se hrává* na více vítězných her.
 match REFL play.GEN.3SG.PRS on more games
 'The match tends to be played in several winning sets.'
- b. [...] z komentáře k Iuliovu zákonu de maiestate z Digest
 from commentary to Iulius law de maiestate from digest
 císaře Justiniana: [...] 4) kdo *svolává* k
 emperor Justinian: 4) who convene.GEN.3SG.PRS to
 povstání, [...] insurrection
 '[...] from the commentary to Iulius' law de maiestate from the Digest of the Emperor Justinian: [...] who tends to call for an insurrection [...]

For such rules and regulations, using a verb with the generic morpheme is useful as a hedging device. Due to its incompatibility with exceptionless generalizations, the generic morpheme explicitly assures a certain leeway in the application of a rule of law, and hedges against unprecedented or unexpected exceptions to it. In contrast, rules and regulations, such as those above, if they were formulated with morphologically related non-generic verbs (without the generic morpheme *-va-*), would be merely exception tolerating; unless the context explicitly indicated otherwise, they would likely be pragmatically strengthened to strict, exceptionless rules.

³³Example (44a) is from <https://cs.wikipedia.org/wiki/Kuličky>, accessed on April 9, 2022. Example (44b) is from https://is.muni.cz/th/nmkho/Frydek_-_Crimen_maiestatis_-_final.pdf; accessed on August 16, 2025.

The use of the generic marker *-va-* in generics describing various rules and norms fits Carlson's (1995) rules-and-regulations model for the truth conditions of generic sentences (see Section 2 above). Recall that according to this model, generic sentences are true by virtue of some underlying rule, principle or cause, which we may directly learn, rather than (just) by virtue of real world particulars from which we induce the generalization. The distribution of the generic marker *-va-* cuts across the divide between the inductive and rules-and-regulations model, and also across the descriptive/normative distinction. It clearly is not a marker of only inductive generalizations, including habits proper, which are paradigm examples for the inductive model (Carlson 1995). However, the rules and regulations that are expressed by generic sentences formally marked with *-va-* must have been already instantiated, which follows from its realization requirement (see Semantic Property 1 in (19)).

4.2 The quantificational properties of the generic morpheme *-va-*

4.2.1 Quantification over situations and/or individuals

One common way of representing the logical structure of generic sentences was introduced in Section 2: namely, the tripartite structure with GEN, which is a phonologically null, modalized dyadic operator. The tripartite structure was originally proposed for the logical structure of sentences with overt Q-adverbs (Lewis 1975). As Chierchia (1995) observes, GEN patterns with overt Q-adverbs when it comes to its variable-binding properties:

- (45) Variable-binding properties of the generic operator GEN (see e.g., Chierchia 1995):
- a. a situation variable (in generic sentences without kind terms)
 - b. variables provided by singular indefinites and bare plurals
 - c. variables provided by kind-denoting definites
 - d. more than one variable (in generic sentences with kind terms)
 - e. relatively freely select the arguments for its binding, modulo context

The generic marker *-va-* appears to share its variable binding properties with the generic operator GEN (see also Filip 2009, 2021), which serves as one among other arguments given here in support of its status as a generic marker. To illustrate this point, I will adduce some examples from Czech. Comparable uses of this morpheme in other Slavic languages that have analogous verbs with the generic marker will exhibit the same binding properties, as can be easily verified.

The representational style of the simplified tripartite structures that are given below follows Krifka et al. (1995) and Pelletier & Asher (1997) (see also above in

Section 2). The contribution of the generic morpheme *-va-* is represented in the following tripartite structures by means of the quantifier $\forall A$, a kind of generic quantifier.^{34,35}

- (46) BINDING OF SITUATION VARIABLES in generics without kind terms
- Po práci si Tom dávává pivo.
after work REFL Tom give.GEN.3SG.PRS beer
'After work, Tom has (a) beer.' [i.e., now and then / rarely / often ...]
 - $\forall A[s]; (TOM \text{ in } s \wedge s \text{ AFTER WORK}; TOM \text{ HAS (A) BEER in } s)$
'For all appropriate after-work situations s such that Tom is in s , Tom has (a) beer in s .'
- (47) BINDING OF VARIABLES PROVIDED BY INDEFINITES
- Židle {mívá / mívají} čtyři nohy.
chair(s) have.GEN.3SG.PRS have.GEN.3PL.PRS four legs
'A typical chair/Chairs tend(s) to have four legs.'
 - $\forall A[x]; (CHAIR(x); HAVE \text{ FOUR LEGS}(x))$
'When a thing has the property of being a chair, it has four legs.'
'For a given thing x such that x is a chair, x has four legs.'
- (48) BINDING OF VARIABLES PROVIDED BY KIND-DENOTING TERMS
- Člověk se k stáru měnívá.
man REFL toward old.age change.GEN.3SG.PRS
'Man changes as he grows old.' (from K. Čapek: *Obyčejný život*)
 - $\forall A[x, s]; (MAN(x) \wedge x \text{ in } s \wedge s \text{ OLD AGE}; x \text{ CHANGE in } s)$

³⁴The treatment of the proper name *Tom* in the formula (46b) follows the treatment of the proper name *Fred* in *Fred always smokes* in Chierchia (1995: 189, ex. (37)). It is treated as a constant in the logical representation of *Fred always smokes*; the universal Q-adverb *always* only binds the situation variable; its variable binding properties, according to Chierchia and others, pattern with GEN. However, strictly speaking, the generic operator GEN in this type of generic sentence quantifies over different stages of the name bearer participating in different corresponding particular situations in which the characterizing property is manifested. This is implied in an alternative tripartite structure associated with this type of sentence in Krifka et al. (1995), where *Tom* introduces the individual variable x , which is satisfied by different stages of Tom at particular situations: *Tom drinks beer after work* $\rightarrow \text{GEN}[x, s](x = TOM \wedge x \text{ in } s \wedge s \text{ AFTER WORK}; x \text{ DRINK_BEER in } s)$.

³⁵Slavic languages lack an article system; only South Slavic languages like Bulgarian and Macedonian have what is taken to be a definite article. The bare subject argument in the following Czech examples can be viewed as corresponding to the English singular definite subject, bare plural subject, and where the predicate does not denote a contingent property also to the English indefinite subject argument.

(49) BINDING OF MORE THAN ONE INDIVIDUAL VARIABLE

- a. Kočka honívá myš.
 cat chase.GEN.3SG.PRS mouse
 ‘Cats tend to chase mice.’
- b. $\forall x[s, x, y;](\text{CAT}(x) \wedge \text{MOUSE}(y) \wedge x \text{ in } s \wedge y \text{ in } s; x \text{ CHASE } y \text{ in } s)$

As we see above, the Czech marker *-va-*, which introduces the quantifier $\forall$ into the tripartite structure, semantically functions as a quantifier over (i) situations, as in (46a), (ii) individuals, as in (47a), and (iii) both situations and individuals in sentences with kind denoting terms, as in (48a) and (49a), which is aligned with the quantificational properties of the null generic operator GEN.

In (47a), the Czech marker *-va-* functions as a quantifier whose range only contains individual members of a kind. This invalidates its treatment as an iterative marker denoting a function that returns a set of plural dynamic situations of the same type, at a given world and time. Examples like (47a) clearly show that the standard designations of *-va-* as an iterative, multiplicative, or frequentative marker (e.g., Petr 1986, Kosek 2014, Lamprecht et al. 1986, Dahl 1995) are misleading, if not wrong.

Moreover, examples like (47a) invalidate the claim made by Dahl (1995: 421) that the Czech marker *-va-* serves “as a kind of quantifier over situations with, roughly, the semantics of ‘most’”. Following the terminology of Dahl 1985, let us refer to this kind of tense-aspect markers as *habituals*.” Given that this claim is false, and given that it is one of the two key claims that Dahl uses to deny that the Czech morpheme *-va-* has a status of a generic marker, there really remains no valid reason, which Dahl adduces, why *-va-* cannot be analyzed as a generic marker *sui generis*, given that the other argument, namely the obligatory occurrence in all and only generic sentences is also spurious. Given that Dahl uses the Czech marker *-va-* as a paradigm example of comparable markers in other typologically distinct languages (see Dahl 1995: 421, fn. 8) that enforce a generic (and habitual) interpretation, this also raises at least some doubts about their purported status as “tense-aspect habitual markers”.

The use of the generic morpheme *-va-* in examples like (47a) also shows that it is clearly not a “habitual” morpheme in the proper sense of “habit”, i.e., dedicated to formally marking only sentences that quantify over situations with animate individuals, typically human agents, as in (46a). Another example of a generic sentence marked with *-va-* that has nothing to do with habits in the narrow sense is given below:

- (50) Moře tady bývá v tuto dobu vyhřáté na příjemných 28° C.
 sea here be.GEN.PRS in this time warmed-up on pleasant 28° C
 ‘At this time the sea here tends to be warmed up to a pleasant 28° C.’

To illustrate the breadth of use of the Czech generic morpheme, we may also mention that it commonly occurs in generic sentences with dedicated kind-denoting terms like ‘man’, as in (48a). It also occurs in kind reference sentences (see Section 2) that contain kind-level predicates like *be widespread* directly selecting kind-denoting terms for one of their arguments and which assign characterizing properties to kinds that can never hold of its individual members, as we see in (51) below:

- (51) Rohozub nachový bývá rozšířený u lidských sídlišť.
 ceratodon purpureus be.GEN.3SG.PRS widespread at human dwellings
 ‘The ceratodon purpureus tends to be widespread close to human dwellings.’

There is a growing body of studies on markers that, just like the Slavic generic marker, constitute a sufficient, but not a necessary, condition for the expression of various regularities, and often, similarly as in Dahl (1995), they are treated as “habitual”, rather than generic, markers. Some examples are: the Tlingit (Na-Dene; Alaska, British Columbia, Yukon) particle *nooch* (Cable 2022), the *be*-construction in African American English (Green 2000), or the construction with *haya*+participle in Modern Hebrew (Boneh & Doron 2008). It is striking that such studies focus on sentences that describe habits in the proper sense, such as (46a). It then raises the question whether this empirical focus of studies on such “habitual” markers and on habits proper is due to a data elicitation bias, and whether the relevant markers in these languages actually cover a broader swath of data, similar to those examples that we find in Czech. One wonders whether their quantificational properties come close to or perhaps even match those of the null generic operator GEN, or the Czech generic marker *-va-*, and by the same token, whether their treatment as merely “habitual” markers (possibly subsumed under tense and/or aspect) which are claimed to be separate from genericity might not be reconsidered.

4.2.2 The Slavic generic morpheme *-va-* is not an overt Q-adverb: Irreducibility to ‘usually’ or to any other single quantifier or quantity expression

In standard reference grammars of Czech, the meaning of the Czech morpheme *-va-* is often characterized in terms of “usuality” (*uzuálnost* in Czech, see e.g., Petr

1986: 184, i.a.). Hence, Dahl's (1995) claim that the Czech morpheme *-va-* functions as a quantifier over situations with the meaning of approximately 'most' or 'usually' might seem plausible. In Dahl (1995), this semantic claim is based on two rather flimsy data points: namely, "the glosses given in Kučera (1981)" (Dahl 1995: 421), and a cursory observation (Dahl 1995: 420–421) regarding the distribution of the Czech *-va-* and similar markers from other languages over his two generic contexts: namely, (i) it tends to be optional in the context of 'usually', and (ii) it is typically absent in the context that concerns a job description.³⁶

While Dahl (1995) does not motivate the choice of these two generic contexts, they seem well-chosen. The first would seem to elicit generics that are roughly aligned with Carlson's (1995) inductive model of genericity, and concern weak descriptive generalizations, while the second with Carlson's (1995) rules-and-regulations model of genericity. However, as shown above (Section 4.1.3), the Czech generic *-va-* can be used not only in generic sentences that describe weak descriptive generalizations, but also in generics stating a variety of rules (e.g., game rules), including the rules of the law.

Now, if the Czech *-va-* had a meaning of approximately 'most' or 'usually', as Dahl (1995) claims, it would not qualify as a generic morpheme. However, the Czech *-va-* contributes no requirement on a specific quantity of situations or instances that count as evidence for the truth of the generalization. This is a hallmark property of markers of genericity (and habituality as its special case) in natural languages. In this respect, the generic morpheme *-va-* patterns with the phonologically null GEN operator. Consequently, just as GEN, it cannot be reduced to 'most' or 'usually', or the meaning of any other single quantifier or expression of quantity, no matter how modalized (see 2 above). Let me adduce a few concrete examples to illustrate this point.

First, we observe that the Czech marker *-va-* freely co-occurs with virtually any expression of quantification or quantity (see, for instance, corpus studies of Šírková 1963: 62, 81; Šírková 1965, Danaher 2003). Danaher's (2003) corpus study shows that *-va-* occurs less often with *obvykle* 'usually' than with Q-adverb(ial)s denoting a low frequency like *občas* 'from time to time', *někdy* 'sometimes', *málokdy* 'rarely', *tu a tam* 'here and there', *vzácně* 'rarely'.³⁷ Any of these Q-adverb(ial)s can fill the [Q-ADVERB] slot of (52):

³⁶In Dahl (1995: 420–421), the relevant test context is given as follows: [A: My brother works at an office. B: What kind of work he DO? A:] He WRITE letters.

³⁷Danaher's (2003) corpus comprises 376 attested examples of verbs marked with the marker *-va-*, collected from a wide variety of text genres (e.g., contemporary literary essays, fiction, memoirs, journalistic prose and scholarly writing).

- (52) Po snídani Tomáš Q-ADVERB psává dopisy.
 after breakfast Thomas Q-ADVERB write.VA.PRS letters
 ‘Thomas Q-ADVERB writes letters after breakfast.’

Second, *-va-* can be used to formally mark generic sentences that are true even if *most* relevant instances do *not* satisfy the generically-predicated property, as we see in (53), which makes its reductive analysis to ‘most’ or ‘usually’ entirely implausible.³⁸

- (53) Za Putina, ruští oligarchové umírávali pádem z okna.
 during Putin Russian oligarchs die.GEN.3PL.PST fall from window
 ‘In Putin’s times, Russian oligarchs tended to die by falling out of windows.’

(53) is true even if as a matter of fact most Russian oligarchs who died in Putin’s times did not die by falling out of windows, only a very small number did. The reason why we judge this sentence as true does not hinge on the number of Russian oligarchs dying by falling out of windows, but rather on viewing the nature of their death as so highly appalling, striking, unexpected, or memorable that we treat it as their characterizing property. This kind of subjective evaluation is based on the same psychological mechanism that motivates why we judge generic sentences like (9c) *Mosquitoes carry the West Nile virus* as true, even if only about 1 % of mosquitoes carry the virus.

Moreover, adding ‘most’ or ‘usually’ to true generic sentences like (53) that are formally marked with *-va-* results in false sentences, as we see in (54a) and (54b). Given that a generic sentence marked with *-va-* does not entail the corresponding sentence with the added quantifier ‘most’ or ‘usually’, *-va-* does not have as part of its meaning the quantificational import of ‘most’ or ‘usually’.

- (54) a. Za Putina, většina oligarchů umírávala pádem z okna.
 during Putin most oligarchs die.GEN.PST fall from window
 ‘In Putin’s times, most oligarchs died by falling out of windows.’
 (53): TRUE $\nRightarrow$ (54a): FALSE

³⁸This example is tailored on Kučera’s (1981) example:

- (i) Za Stalina ruští generálové umírávali v mladém věku.
 during Stalin Russian generals died.GEN in young age
 ‘In Stalin’s times, Russian generals tended to die young.’

Kučera translates this sentence as ‘Most generals died young in Stalin’s times’, but this does not seem to be correct, given that what it describes is factually false; most Russian generals did not in fact die young in Stalin’s times, as we know.

- b. Za Putina, oligarchové *obyčejně* umírávali pádem z okna.
 during Putin oligarchs usually die.GEN.PST fall from window
 ‘In Putin’s times, oligarchs usually died by falling out of windows.’
 (53): TRUE $\Rightarrow$ (54b): FALSE

Neither can (53) with the generic marker *-va-* be paraphrased by a corresponding sentence with an overt Q-adverb like ‘usually’ and a non-generic imperfective verb without *-va-*. While (53) is true, (55) is false:

- (55) Za Putina, oligarchové *obyčejně* umírali^{IPFV} pádem z okna.
 during Putin oligarchs usually die.PST fall from window
 ‘In Putin’s times, oligarchs usually died by falling out of windows.’
 (53): TRUE $\Rightarrow$ (55): FALSE

In sum, the generic morpheme *-va-* in Czech does not function as a “habitual” marker that quantifies over situations only, and its meaning cannot be reduced to that of an overt Q-adverb like ‘usually’, or any other quantifier or expression of quantity, contrary to Dahl (1995). In Slavic languages that have analogous verbs with the generic marker *-va-*, we can observe the same general interpretation pattern concerning its quantificational import.

4.3 Summary

At the outset, one of the main questions posed concerned the relation of the meaning of the generic morpheme *-va-* to the phonologically null generic GEN operator, which in quantificational theories of genericity is hypothesized to stand for the key meaning component of generic and also habitual sentences (see (11)). Based on the data and observations so far, and assuming that the generic morpheme *-va-* introduces a kind of generic operator VA into the tripartite logical representation, the similarities and differences between VA and GEN can be summarized as follows.

- (56) Shared properties of the generic operator VA (in Czech) and the null generic operator GEN:
- They model the meanings of sentences that only have a generic interpretation (and disallow any episodic interpretation).
 - They derive complex generic predicates that are aspectually stative, one of the signature properties of generic predicates (see above Section 2, and Semantic Property 1 in (25)).

- c. Their analysis cannot be reduced to the meaning of any single quantity expression like *usually*, *generally*, *typically*, *most*, or *in a significant number of cases*, of any extensional quantifier (e.g., universal, existential), or any single, purely quantitative criterion or quantity-based measure, regardless whether they are characterized in modal or probabilistic terms (see Section 4.2.2).
- d. Both are modal generic operators; their meaning does not (just) depend on a mere plurality of particulars, situations or “cases” (in the sense of Lewis 1975), but rather they add predictive force to the meaning of sentences, which has implications for other accessible worlds (perhaps all), than just one possible (actual) world, and supports counterfactual inferences.
- e. They seem to exhibit the same variable binding properties (akin to those of overt Q-adverbs) (see Section 4.2.1).

The generic operator *VA* differs from *GEN* in so far as it adds two uncancellable inferences (as part of its truth-conditional meaning) to the meaning of generic sentences. For convenience, they are repeated here as follows:

(57) Two uncancellable inferences

- a. Exceptive inference: There are actual (known) or possible exceptions or counterexamples to the generalization (Semantic Property 2 in (26)).
- b. Realization inference: There are verifying instances in the actual world that count as evidence for the truth of the generalization (Semantic Property 3 in (38)).

The co-occurrence of these two uncancellable inferences may not be a matter of mere accident. The generic *-va-*, as I argue, semantically functions as a modal generic quantifier *sui generis* whose use involves reasoning with exceptions and counterexamples; it involves the speaker’s (implicit) background understanding of what counts as salient reasons for/against the correctness of applying the generalization, in the light of exceptions or counterexamples which the speaker thinks cannot be safely ignored or tolerated, out of abundance of caution, to ensure plausible deniability and the like. This kind of reasoning with exceptions and counterexamples is motivated by the speaker’s more or less direct experience of the facts, particulars in the actual world, of how various states of affairs are realized in the *actual* world. If so, then it should not be surprising that *-va-* also carries the uncancellable inference that there are verifying instances in the

actual world that count as evidence for the truth of generic sentences (realization inference).

Due to these two uncancellable inferences, the Slavic generic morpheme *-va-* delimits only a subset of generic sentences that the null generic operator GEN underpins; GEN is introduced into the logical representation of generics that not only sanction exceptions, but also of generics that (necessarily) hold with no exceptions like the universal laws of nature, and also purely hypothetical generics that are true even if there have been no instances that count as evidence for their truth, and may never be.

As in English, the null generic operator GEN may stand for a representation of the generic (and habitual) meaning in the logical structure of Slavic generic (and habitual) sentences that are headed by verbs that are *not* marked for genericity, i.e., non-generic verbs (see 1): these are either primary imperfective verbs (formally unmarked for imperfectivity) or secondary imperfectives (which are marked for imperfectivity with the imperfectivizing suffix), and also perfective verbs, in particular in West Slavic languages where they are commonly used in generic (and habitual) sentences.

5 Proposal

5.1 Reasoning with exceptions

One key difference between generic sentences formally marked with *-va-* and those that lack this marker has to do with the STRENGTH of the generalization they express which is calibrated in terms of the speaker's (public) commitment to the actual or possible exceptions or counterexamples that the generalization *may*, *must* or *must not* have.³⁹ (These three salient cases were discussed in Section 4.1.)

Intuitively, the strongest generalizations have no exceptions or counterexamples whatsoever in every possible world/in all the known circumstances (e.g., universal laws of nature *The Earth revolves around the Sun* or definitional statements like *A triangle has three sides*). Lower on this scale are non-universal generalizations (including laws and rules that fall under special sciences such as psychology, biology, and economics, e.g., *As the price of a good or service increases, the quantity demanded decreases, and vice versa.*) which in principle allow for ex-

³⁹ Assuming that making a statement publicly commits the speaker to its content: A "speaker is committed to a statement when he makes it himself, or agrees to it as made by someone else, or if he makes or agrees to other statements from which it clearly follows" (Hamblin 1971, Hamblin 1987: 136).

ceptions, but may be compatible with no exceptions, followed by generalizations that have (known) exceptions that we tolerate (e.g. *Dogs bark*).

For all these types of generalizations, in Czech there are freely available non-generic (im)perfective verb forms (i.e., forms without the generic *-va-*), unless, they have exceptions that cannot be tolerated (positive counterinstances), in which case, just as in English, some overt expression is needed to qualify the generalization and ensure its truth. By using a non-generic verb in a non-universal generic sentence, which in principle allows for exceptions, the speaker may remain publicly non-committal as to whether there are exceptions to the expressed generalization, but also, in an appropriate context and the nature of the generalization, allow for pragmatic strengthening to an exceptionless generalization.

In contrast, the use of a specifically generic verb with the generic marker *-va-* commits the speaker to there being actual exceptions or counterexamples, or possible exceptions to the generalization that the speaker cannot categorically exclude (due to the lack of sufficient information) in every possible world (see Semantic Property 2 in (26)). This means that the generic morpheme *-va-* can only be used to mark generic sentences that express non-universal generalizations, and which exclude pragmatic strengthening to an exceptionless generalization. It may be used as a hedge to safeguard a generalization against refutation by states of affairs of which the speaker is ignorant, and so to ensure its plausible deniability. Such generalizations are akin to generalizations with exceptives like *but*, prepositions *except (for)*, or with a *ceteris paribus* clause.

The difference in the strength of the generalization expressed by generic sentences marked with *-va-*, on the one hand, and of the generalization expressed by formally unmarked generics, on the other hand, can be represented, as I propose, in terms of a *rank order* (Lehrer 1974; Hirschberg 1985; Geurts 2010; Horn 1989, 2000, 2010, 2017, i.a.). Rank orders, which have the general format of $\langle\langle\text{strong} \mid \text{weak}\rangle\rangle$, are related to, but distinct from, scales for entailment-based scalar implicature. Both rank-order based and entailment based scalar implicatures are a kind of quantity implicature. However, in rank order based implicatures, stronger elements do not entail weaker ones, but rather we can infer their negations:

(58) SCALE ORDER VERSUS RANK ORDER (Horn 2017)

SCALE: In a scale of the form $\langle Y, X \rangle$, ... *Y* ... unilaterally entails ... *X* ...

Example: $\langle \textit{all}, \textit{most}, \textit{many}, \textit{some} \rangle$

RANK ORDER: In a rank order of the form $\langle\langle Y \mid X \rangle\rangle$, ... *Y* ... unilaterally entails ... $\neg X$...

Examples:

$\langle\langle married \mid engaged \rangle\rangle$: ‘if a person is married, it is false that that person is engaged (to be married)’

$\langle\langle dead \mid sick \rangle\rangle$: ‘if a person is dead, it is false that that person is sick’

Unlike values within a scale, ranked elements are mutually incompatible. For instance, being married does not entail being engaged. Applied to generic sentences with and without the generic morpheme *-va-* in Slavic languages, we get the following rank order:

- (59) RANK ORDER (preliminary version): $\langle\langle [GEN]P \mid [VA]P \rangle\rangle$

where,

$[GEN]P$: compatible with no exceptions

$[VA]P$: incompatible with no exceptions

In a rank order $\langle\langle [GEN]P \mid [VA]P \rangle\rangle$, $[GEN]P$ unilaterally entails $\neg([VA]P)$, all else being equal.

The rank order in (59) is based on two main lexical alternatives that Slavic languages have available for the expression of generic statements: namely, (i) verbs that are formally marked for genericity with the generic morpheme *-va-*, which introduces the generic operator *VA* into the logical representation and here represented with *VA* as part of the weak element of the rank order, and (ii) non-generic verbs without the generic morpheme *-va-*, which have contextually determined generic uses, and occur in generic sentences that introduce the null generic operator *GEN* into the logical representation, and here represented with *GEN* as part of the strong element of the rank order. The two ranked alternatives, strong and weak, are equally informative, and do not stand in an entailment relation to each other: namely, the strong element does not a fortiori imply the truth of the weak one, but rather from the strong element we can infer the negation of the weak element. Let me illustrate this with the following contrast:

- (60) a. Petra u nás řídí^{IPFV} autobus.
 Petra at us drive.3SG.PRS bus
 ‘Petra drives a bus here [i.e., is a bus driver].’
- b. Petra u nás řídívá autobus, když je nedostatek
 Petra at us drive.GEN.3SG.PST bus when be.3SG.PST shortage
 řidičů.
 driver.MASC.GEN.PL
 ‘Petra tends to drive a bus here [i.e., drives a bus now and then, with some regularity], when there’s a shortage of bus drivers.’

Unless the context indicates otherwise, from (60a) we may naturally infer that Petra is employed as a bus driver, being a bus driver is her profession. As such it is her characterizing property that holds of her stably over a large chunk of her life-time, i.e., at all its subintervals and moments of time without interruptions, including times when she happens to be off work, e.g., on vacation or sick. In contrast, (60b) is naturally understood as meaning that Petra drives a bus from time to time, often, occasionally, just in case the appropriate occasion arises, and the like. Hence, (60a) does not entail (60b), but rather from (60a) we can infer the negation of (60b). And vice versa, (60b) does not entail (60a). This can be also confirmed by the observation that in (61), B's cancellation with 'yes' is odd or at least dispreferred:

- (61) A: Řídívá Petra u Vás autobus?
 drive.GEN.3SG.PRS Petra at you bus
 'Does Petra {occasionally / from time to time / often...} drive a bus in
 your city?'
 B: {Ne / #Ano}, Petra u nás (dokonce) řídí^{IPFV} autobus.
 no yes Petra at us even drive.3SG.PRS bus
 'No, (well) Petra (in fact) is a bus driver here/in our city.'
 '#Yes, Petra (in fact) is a bus driver here/in our city.'

Rank orders, similarly to scale orders, give rise to quantity implicatures, which are inferences triggered by an utterance and the alternative utterance(s) a speaker could have made instead. The rank order (59) may trigger one of the two types of inference, depending on context: certainty or ignorance regarding the speaker's commitment to exceptions. The choice of the non-generic alternative, i.e., the strong element in the rank order in (59), may in an appropriate context convey a certainty inference (by pragmatic strengthening) that there are *no exceptions* to the generalization in all the known conditions, in all the speaker's doxastic alternatives. This constitutes the strongest, upper bound of the rank order.

The use of the generic morpheme *-va-*, which corresponds to the weak alternative in the rank order (59), signals that there are exceptions or counterexamples to the generalization that are of the kind that the speaker thinks cannot be safely ignored, given the speaker's (implicit) understanding of the correct application of the generalization, weighing some salient reasons for or against it. The use of *-va-* triggers one of the two inferences, depending on context. The speaker wishes to convey the inference that they know that there are exceptions or counterexamples (certainty inference), *or* the speaker does not have enough information to be in the position to exclude them (ignorance inference).

Let us examine these two inferences in more detail. By using the generic *-va-*verb (the weak alternative) when the verb without the generic morpheme (the strong alternative) could have also been used (*salva veritate*), the speaker may convey the ignorance inference that they are not in the position to use the strong alternative, because it allows pragmatic strengthening to an exceptionless generalization, unless it is explicitly indicated otherwise, and the hearer might be misled. This case has been illustrated with the minimal pair in (43) above, repeated below.

- (62) a. Výuka začíná^{IPFV} od 8.00 a končí v poledne.
 teaching start.3SG.PRS from 8:00 and ends in noon
 ‘Classes start at 8 a.m. and end at noon.’
 b. Výuka začínává od 8.00 a končí v poledne.
 teaching starts.GEN.3SG.PRS from 8:00 and ends in noon
 ‘Classes tend to start at 8 a.m. and end at noon.’ [as a rule, usually]

Let us suppose there is a subset of (62a)’s speaker’s doxastic alternatives where the school opens at 8 a.m. and a subset where it does not; in other words, the speaker of (62a) does not know which of these alternatives holds, and considers both as live possibilities (ignorance inference). Uttering (62b) with the generic morpheme, rather than (62a) without it, readily allows for plausible deniability in the face of refutation by information of which the speaker of (62a) is ignorant, and specifically should it turn out that the school does not always open at 8 a.m. In contrast, by uttering (62a) with the non-generic verb, rather than (62b) with the generic one, entails the negation of the latter, and, provided the context does not indicate otherwise, naturally leads to a pragmatic strengthening of (62a) to an exceptionless generalization. I.e., (62a) is compatible with the certainty inference that in all the speaker’s doxastic alternatives associated with (62a) the school opens at 8 a.m.

The exceptive inference of the generic verb marked with *-va-* (see Semantic Property 2 in (26)) can also be exploited to convey the certainty inference that there are counterexamples to the generalization that would have made the corresponding sentence without *-va-* false, all else being equal. The best examples are generalizations with counterexamples that count as positive counterinstances (in the sense of Leslie 2008), such as hardbacks that falsify *Books are paperbacks*. These are inadmissible non-confirming cases which *cannot* be (easily) ignored when judging the truth of the generalization. Above, this was illustrated by (42) and the relevant contrast is repeated below. By using the generic morpheme in (63b), as opposed to the alternative without *-va-* in (63a), the speaker (publicly)

commits to the existence of counterexamples to the generalization, which, in turn, prohibits pragmatic strengthening to an exceptionless generalization, and guarantees the truth of (63b).

- (63) a. *Knihy jsou^{IPFV} brožované.* FALSE
 books be.3PL.PRS paperback
 ‘Books are paperback.’
 b. *Knihy bývají brožované.* TRUE
 books be.GEN.3PL.PRS paperback
 ‘Books tend to be paperback.’

In representing such speaker-oriented certainty and ignorance inferences, we may rely on the covert doxastic operator $\Box_{SP(c)}$ (inspired by Meyer 2013), which is understood as universally quantifying over the speaker’s doxastic alternatives (Degano et al. 2025):

- (64) A covert doxastic operator: ‘ $\Box_{SP(c)}(P)$ ’ where,
 ‘ $SP(c)$ ’: a doxastic source, the speaker in context *c*
 ‘ $\Box_{SP(c)}$ ’: a covert doxastic operator
 ‘ $\Box_{SP(c)}(P)$ ’: in all the speaker’s doxastic alternatives, the speaker knows that *P*

Specifically for the representation of the ignorance inference, we may employ the classic definition of ignorance of Hintikka (1962), characterized as ‘not knowing whether’. In our case, and assuming that *P* corresponds to the strongest value in the rank order of the form $\langle\langle\text{Strong} \mid \text{Weak}\rangle\rangle$ and compatible with “all exceptions categorically excluded”, we have: (i) the speaker does not know whether *P* is true, (ii) the speaker does not know whether $\neg P$ is true, and (iii) the speaker also considers both *P* and $\neg P$ as live possibilities compatible with the speaker’s knowledge.⁴⁰

With these ingredients in place, adding the two speaker-oriented inferences to our original rank order (59), its final formulation is given below.

⁴⁰Traditionally, and based on Grice (e.g., Grice 1975, 1989, i.a.), ignorance inferences are derived by means of maxims of conversation (quantity and quality), based on general principles of conversation and cooperative behavior of interlocutors. Ignorance inferences are taken to consist of uncertainty and possibility inferences, where the possibility inference is derived from uncertainty. Recently, however, Degano et al. (2025) have argued for a possibility inference without the uncertainty inference. Due the limits on this chapter, I cannot here address the debates concerning ignorance inferences in more detail.

(65) RANK ORDER (final version): $\langle\langle [\text{GEN}]P \mid [\text{VA}]P \rangle\rangle$

where,

$[\text{GEN}]P$: compatible with no exceptions

$[\text{VA}]P$: incompatible with no exceptions

- (i) Asserting that $[\text{GEN}]P$ holds, given that the speaker could have instead asserted that $[\text{VA}]P$ holds, may trigger

CERTAINTY INFERENCE: $\Box_{\text{SP}(c)}(P)$

unless the context indicates otherwise.

- (ii) Asserting that $[\text{VA}]P$ holds, given that the speaker could have instead asserted that $[\text{GEN}]P$ holds, triggers one of the following inferences depending on context:

(a) CERTAINTY INFERENCE: $\Box_{\text{SP}(c)}\neg(P)$

(b) IGNORANCE INFERENCE:

$$\neg\Box_{\text{SP}(c)}(P) \wedge \neg\Box_{\text{SP}(c)}\neg(P) \leftrightarrow \Diamond_{\text{SP}(c)}(P) \wedge \Diamond_{\text{SP}(c)}\neg(P)$$

(following Hintikka 1962)

In effect, generic sentences formally marked with the generic morpheme *-va-* are endowed with two modal (intensional) components: one that is tied to their generic, law-like, or gnomic force, which it shares with the null generic operator GEN, and the other which involves reasoning with exceptions and counterexamples, which may be characterized in terms of the speaker-oriented doxastic (and epistemic) modality, in interaction with general pragmatic principles of interpretation.⁴¹

5.2 Supporting Evidence: The *va* operator with overt restrictors

The properties of the generic marker *-va-* are relevant to discussions of generic sentences in the context of *ceteris paribus* (CP) generalizations, given that, as

⁴¹The speaker-oriented modality that concerns the speaker's commitment to the strength of the generalization, which is based on the speaker's information state about exceptions or counterexamples to it can be seen as related to observations in the generic literature that generic sentences track not only (relatively) objective facts about the world, but also a variety of subjective factors, including the speaker's information state, beliefs about the facts of the world, and a variety of attitudes that the speaker may take towards them. As has been observed above, some generics track what we consider to be particularly salient, striking, dangerous or impressive facts (e.g., *Sharks attack swimmers*), which, for instance, Leslie (2008) integrates into her cognitive approach to generic sentences, or they track psychologically salient partitions of the kind (Cohen 2004), or reflect judgments based on stereotypes (e.g., *Asian people are good at mathematics*).

I propose, its use is driven by reasoning with exceptions, counterexamples, or various restricting conditions which cannot be ignored or tolerated. Philosophers draw on generic sentences in order to advance our understanding of the nature and status of CP generalizations in special sciences (see e.g. Nickel 2016 and references therein). A line is drawn between exceptions to the CP generalization that are compatible with its truth and counterexamples which are not.

The distribution of *-va-* over different types of generic sentences taps into this distinction, as has been illustrated by a number of examples above. To what has already been observed, we may add that in contrast to non-generic verbs, there are contexts in which specifically generic verbs marked with *-va-* may be odd or dispreferred with no overt restrictor, but perfectly acceptable if they are licensed by an appropriate overt restrictor. This can be shown in generic sentences with kind reference that describe what are taken to be characterizing properties of a kind to which there are tolerable exceptions compatible with their truth, but which would refute the corresponding universal generalization. Consider the following contrast:

- (66) EXCEPTIONS: non-barking dogs (negative counterinstances, see Leslie 2008)

Context: ‘What sound do dogs make?’

- a. Psi štěkají^{IPFV}. [=(1a)]
 dogs bark.3PL.PRS
 ‘Dogs bark.’
- b. (#) Psi štěkávají.
 dogs bark.GEN.3PL.PRS
 ‘Dogs tend to bark [when... / if... / other things being equal / unless prevented].’

In this context, the preferred and the most natural answer is (66a) with the verb without *-va-*. (66a) and its closest English correspondent in this context *Dogs bark* instantiates a prototypical generic sentence (see Section 2): namely, they have no overt restrictor or quantificational element, their bare plural subject term denotes a kind whose characterizing property is denoted by a verb form that is the least marked form in their respective aspect and tense system, i.e., they exhibit the minimal marking tendency of genericity with respect to tense and aspect (Dahl 1995).⁴²

⁴²In languages with tense and/or aspect systems, prototypical generic sentences exhibit the “minimal marking tendency” of genericity with respect to tense and aspect, as Dahl (1995)

Both (66a) and *Dogs bark* are expected and felicitous answers to the question at hand, because it calls for the basic information about one particular characterizing property of dogs that can be *truly* predicated of dogs as a whole kind, based on our shared kind-related expectations about what it means to be a (realistically) possible dog, by default, typically, or generally, in any possible world; if something is a dog, it will likely bark. Negative counterinstances (in the sense of Leslie 2008), i.e. dogs failing to bark (e.g., breeds like greyhounds and individual dogs that do not bark for a variety of reasons like an injury), and all kinds of disturbing or facilitating factors which may be potentially infinite, are largely irrelevant in this context.

Contrary to this, by using *-va-* in (66b) the speaker shifts focus to exceptions, counterexamples, (possible) restricting conditions on the application of the generalization, implying that (i) members of the kind *DOG* are not homogeneous with respect to the characterizing property of barking, or (ii) do not bark in all the appropriate situations in which not exceptional barking dogs might be expected to bark. This straightforwardly follows from Semantic Property 2 in (26). By using the marked alternative (66b) with *-va-*, when (66a) without it is preferred as an answer to ‘What sound do dogs make?’, the speaker might be (mis)understood as conveying the ignorance inference that they are uncertain whether barking is a *characterizing* property of dogs. But this would be odd, because it would be inconsistent with our general knowledge that barking is a *characterizing* property of dogs, and the speaker is expected to know this, as well as that there exceptional dogs that fail to bark (negative counterinstances), which we tend to ignore.

Or, and most likely, the speaker, by using the generic *-va-*, may convey the certainty inference that they know that there are other exceptions (apart from the tolerable negative counterinstances which we set aside) that are noteworthy, relevant, striking or possibly not known to the hearer in a given context, which cannot be ignored. And so the speaker intends to draw attention to such exceptional individuals, perhaps to show off his knowledge of dogs. But if so, it is odd for the speaker not to explicitly specify what these exceptions are and why they are worthy of being highlighted. This is also the reason, I submit, why (66b) without an overt restrictor, and out of the blue, is odd or dispreferred.

puts it: namely, their main verb is either devoid of any tense and aspect marking (e.g., *bark* in *Dogs bark*) or corresponds to the least marked form in the verb system. In Slavic languages it is the primary (im)perfective verb; it is a verb form with no derivational morphemes, overt aspect marking, but possibly only marked with tense inflectional morphology. The Czech sentence (66a) on its generic interpretation is headed by the imperfective simplex *štěkají* ‘bark.3PL.PRS’, which is marked for tense, number and person, but has no marker of imperfective aspect.

Once an overt appropriate restrictor is added to the unrestricted (66b), we get a restricted generic sentence in which the generic marker *-va-* is fully felicitous. With an overt restrictor, the generic verb marked with *-va-* and its unmarked counterpart without *-va-* are substitutable *salva veritate*:

- (67) a. Psi {štěkají^{IPFV} / štěkávají}, s výjimkou některých
 dogs bark.3PL.PRS bark.GEN.3PL.PRS with exception some
 druhů psů jako chrti.
 breeds dogs like greyhounds
 ‘Dogs tend to bark, except for some dog breeds like greyhounds.’
 b. Psi {štěkají^{IPFV} / štěkávají} na ty, které neznají.
 dogs bark.3PL.PRS bark.GEN.3PL.PRS at those whom NEG.know
 ‘Dogs bark at those whom they don’t know.’

In (67a) the overt restrictor specifies a subkind of dogs, greyhounds, that are explicitly exempt from the generalization. In (67b), the restrictor constrains a subset of situations in which stages of prototypical, unexceptional dogs generally bark.

These two types of overt qualifications are to be expected, if we assume that the generic morpheme *-va-* functions as a kind of generic quantifier, and generic sentences with kind reference involve a “double generalization” (in terms of Carlson 2011). For instance, *Dogs bark* is true by virtue of (i) *individual dogs* that realize the kind DOG having the disposition of barking (unless they belong to a non-barking kind like greyhounds), and also by virtue of (ii) *particular situations* of barking by a stage of an individual dog.

On the quantificational theory of generic sentences (see Section 2), the relevant quantifier in their logical form is either the covert GEN operator or an overt generic Q-adverb (e.g., *usually*, *typically*). And also, as I argue, the generic VA operator, introduced by the generic marker *-va-*. All three – GEN, a Q-Adverb, and VA – have either a covert restrictor or an overt restrictor.⁴³ Overt restrictors of generic quantifiers constitute a large and a heterogeneous class which includes time/place adverbials (e.g., *in/every winter*), conditional *if/when*-clauses, restrictive modifiers, speaker-oriented epistemics concerning the degree of commitment to the generalization (e.g., *in my opinion*), exceptive constructions (*except/but/besides/apart from X*), and a variety of *ceteris paribus* clauses that specify (causally) disturbing or facilitating factors.

⁴³When there is no overt restrictor, as in (66a) *Dogs bark*, the context is standardly assumed to restrict the quantification to contextually appropriate situations. Roughly, (66a) may then be paraphrased as ‘Generally, for situations *C(s)* that are appropriate for dogs to bark, dogs bark in *s*’.

As we have also seen, the generic marker *-va-* has perfectly acceptable and felicitous uses in sentences with no overt restrictor. If so, the speaker uses it to either ensure the truth of a generic sentence in the face of counterexamples that would falsify the corresponding sentence without it, all else being equal (see Section 4.1.2), or to ensure plausible deniability by preempting pragmatic strengthening to universal/exceptionless generalization (see Section 4.1.3). A case in point illustrating this use of *-va-* is (62b). Here, *-va-* may be used to convey one of the two inferences depending on context. *Either* the speaker conveys the certainty inference that they know that there are counterexamples to the generalization that would refute the corresponding exceptionless generalization, *or* the ignorance inference that they lack enough information for making a stronger claim corresponding to an exceptionless generalization, e.g., they have incomplete beliefs about various possible disturbing factors that may refute that stronger claim.

When the generic marker *-va-* has perfectly acceptable and felicitous uses in sentences with no overt restrictor, it might be thought of as assuming a function that is comparable to that of a CP-condition, besides introducing a generic quantifier. This property clearly sets *-va-* apart from both the covert GEN operator, and also from overt Q-adverbs. The term *ceteris paribus* covers different kinds of phenomena, and has different interpretations. Here, *-va-* may be thought of as roughly amounting to an exclusive and indefinite (or indeterminate) CP-condition, e.g., “other things being equal”, “unless prevented”, “in the absence of disturbing factors”, and the like. The exclusion-clause, as Cartwright (1983) suggests, may always be reformulated as a clause which requires that certain truth conditions for the exclusive CP law hold, namely those which exclude disturbing factors: “the literal translation is ‘other things being equal’, yet it would be more apt to read ‘ceteris paribus’ as ‘other things being right’” (Cartwright 1983: 45). Generally, the number of possible disturbing or enabling factors is potentially infinite, and therefore, the restricting condition in application of the generalization cannot always be strictly and fully specified by means of an overt restrictor.

6 Conclusion and further implications

The Czech marker *-va-*, which, as all agree, is separate from the imperfectivizing suffix, is, as I argue, best treated as a marker of genericity (*pace* e.g., Dahl 1995). It is not a marker of iterativity, frequentativity, multiplicativity, in short a mere plurality of situations, or habituality, and as such a marker that is subsumed under tense and/or imperfective aspect, contrary to what is commonly assumed. One of the *prima facie* reasons why it is wrong to classify and label it as an

iterative marker is the simple fact that it is incompatible with iterative adverbials like ‘three times’, used to count particular episodic situations that are not a part of a larger pattern.

It has been shown that this marker has properties that prohibit its subsumption under tense and that do not neatly fit under imperfectivity, either in Slavic languages or as it is generally understood cross-linguistically (see e.g., Comrie 1976). Neither can its meaning be reduced to ‘most’ or ‘usually’ (contrary to Dahl (1995), i.a.), because it imposes no requirement on a specific quantity of situations or instances counting as evidence for the truth of the generalization. By the same token, its meaning cannot be reduced to any other single quantifier or expression of quantity no matter how modalized or probabilistic it might be.

Among the novel key insights of this study is that the marker *-va-* introduces a generic quantifier into the logical representation of sentences, which, similarly to the null generic operator GEN, has a modal (intensional) import that is tied to its predictive, law-like, force, and hence requires access to other possible worlds than just the actual one. Second, it exhibits variable binding properties that are akin to GEN. Third, generic sentences formally marked *-va-* differ from generic sentences without this marker in so far as their truth conditions and felicity depend on the speaker’s reasoning with exceptions. Specifically, they reflect the speaker’s commitment to the strength of the generalization calibrated in terms of exceptions that the speaker believes it may have or has, possibly also including counterexamples that cannot be ignored. This additional modal component may be characterized in terms of epistemic and doxastic modality, and its integration into the meaning of sentences also exploits general pragmatic principles of interpretation.

An exciting domain of study for future research is the exploration of the cognate generic markers *-va-*, their common citation form across Slavic languages (see e.g., Genis et al. 2021, i.a.), in other Slavic languages, namely, in Slovak, where it is also productive, in Polish, where it is less productive, but common enough, and also in other Slavic languages where its occurrences are treated as a part of a lexicalized combination with a verb base, as in Russian, for instance. Hypothesizing that the properties that are here identified for the Czech marker *-va-* may plausibly also characterize its cognates in other Slavic languages would be at least a concrete point of departure for such (comparative) studies.

Turning to cross-linguistic consequences beyond Czech, and Slavic languages, the properties of the Czech marker *-va-* preclude its assimilation to markers that merely encode a plurality of situations (at a given world/time), i.e., markers that are variously labeled as iterative, frequentative, pluractional, habitual, and the like, in traditional descriptive grammars or contemporary linguistic studies.

The empirical and theoretical arguments presented here in support of the claim that the marker *-va-* in Czech, and also in other Slavic languages where it is attested in comparable examples, is a generic marker might serve as a blueprint for their exploration. As observed above, Dahl (1995) after all takes the Czech marker *-va-* as a paradigm example of a whole class of markers in a number of typologically distinct languages that he labels “habitual tense-aspect” markers, and which, according to him, are not generic markers (the list of languages, is given Dahl (1995: 421, fn. 8) and above). To the extent that his claims concerning the Czech marker *-va-* are invalid, as I showed, this raises at least some doubts whether other members of Dahl’s class that the Czech marker *-va-* is supposed to represent are all indeed “habitual tense-aspect” markers, as Dahl claims. In short, we may pose the following question for future research:

- (68) What is the status of so-called “habitual tense-aspect” markers (in terms of Dahl 1995, i.a.) with respect to the categories of the tense and aspect systems? Do they have (at least some of the) formal and semantic properties that have been here identified for the Czech marker *-va-* which make it a plausible candidate for a generic marker?

Markers that are treated as habitual markers in Tlingit (Cable 2022), Modern Hebrew (Boneh & Doron 2010), African-American Vernacular English (Green 2000), just to name a few, are reported to introduce the realization inference just like the Czech generic morpheme *-va-* does. It is an open empirical question whether they also carry the uncancellable inference that the generalization requires (at least the possibility of) exceptions or counterexamples, given that these two meaning components might be seen as related in the case of the Czech generic marker *-va-*. What is also striking is that such previous studies focus on sentences that describe habits in the strict sense, i.e., regularities of action of ordinary animate individuals, mostly human agents, such as *Fred smokes*. It is unclear whether such so-called “habitual” markers can also be used to mark generic sentences that express regularities that have nothing to do with habits proper, just like the Czech marker *-va-* does.

Perhaps future research will show that at least some markers that are referred to as habitual, and also as iterative, frequentative, multiplicative, and the like, are best classified as markers of genericity. If so, it would further bolster the claim that markers of genericity are not as rare as is thought (*pace* Dahl 1995), and invalidate the widespread traditional belief that they do not exist in natural languages (*pace* Leslie 2008, Leslie & Gelmann 2012, Johnston & Leslie 2012, Rhodes et al. 2018, Cohen 2022, Schiller 2023, i.a.) This would put the semantic

episodic/generic distinction on a firmer empirical ground as a semantic distinction with direct grammatical reflexes, in addition to its indirect reflexes in the tense and aspect systems of natural languages, and in a variety of semantic and pragmatic phenomena. Consequently, this would add further arguments for the independence of genericity from the categories of the TMA system in natural languages, and specifically from tense and imperfective aspect. So far such arguments have mainly been of semantic/pragmatic nature, or based on reasoning strategies involved in generalizations which are developed in philosophy, cognitive psychology and computer science (see e.g., Carlson 1977, 1982, Krifka et al. 1995, Pelletier & Asher 1997).

Assuming that the Czech marker *-va-*, and plausibly also the Slavic marker *-va-*, is a generic marker that introduces a generic operator into the logical representation whose properties overlap with, but are not identical to, those of the null generic operator GEN, brings us to the fundamental question in the domain of genericity posed at the outset (see also Carlson 1977, 1995): Can we provide a unified semantic analysis for all generic sentences? In Czech, generic sentences fall into two main types: one formally marked with *-va-*, and the other formally unmarked. In the latter case, and similarly as in English, the generic (and habitual) reading is a function of the combined meanings of the subject and predicate interacting with context, and it can be represented by means of a tripartite logical structure with the GEN operator. Recall that in all Slavic languages, generic sentences that are not formally marked with the generic *-va-* are headed by primary or secondary imperfectives (marked for imperfectivity with the imperfectivizing suffix), and in West Slavic languages also commonly by perfective verbs.

These two main formal means for the expression of characterizing genericity, one formally marked with the morpheme *-va-* for genericity and the other formally unmarked in this respect, are not neatly aligned with the inductive and the rules-and-regulations model of genericity (Carlson 1995), respectively. The generic morpheme *-va-* is naturally used in generic sentences that provide the best examples in support of the inductive model of genericity, namely generics that express descriptive (weak) generalizations, including habits in the strict sense (i.e., regularities of action ascribed to ordinary animate individuals, as in (13)). However, it is important to emphasize that the Czech generic *-va-* is also used in generic sentences that best fit the rules-and-regulations model of genericity, including generics with a normative reading. In short, its use cuts across the inductive versus rules-and-regulations distinction, and also across the related descriptive versus normative distinction.

Tentatively, it may be concluded that this leaves open the question about the feasibility of a unified semantic analysis for all generic sentences in Czech, and perhaps also in other languages that have a dedicated marker of genericity that is comparable to that in Czech.

Abbreviations

1	first person	NEG	negation
2	second person	NOM	nominative
3	third person	PFV	perfective
ACC	accusative	PL	plural
AUX	auxiliary	PREF	prefix
CP	ceteris paribus	PRS	present tense
DAT	dative	PST	past tense
GEN	generic	REFL	reflexive
INF	infinitive	SG	singular
IPFV	imperfective	VA	morpheme -va-
NCI	negative concord item		

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